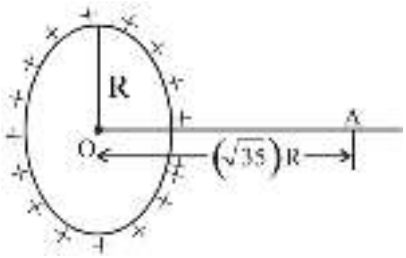
PHYSICS



1) A ring has total charge Q. Find work done by external agent to move –Q charge from point A to centre of ring :-

(1) $\frac{5KQ^2}{6R}$

(2) $\frac{-5}{6}\frac{kQ^2}{R}$

(3) $\frac{3}{2}\frac{kQ^2}{R}$

(4) $\frac{kQ^2}{4R}$

2) The number of electrons, removed from a spherical conductor (radius = 0.1m) to produce 0.36 N/C electric field at the surface of this is :-

(1) 2.5×10^7

(2) 2.5×10^6

(3) 2.5×10^8

(4) 2.5×10^5

3) The electric potential at a point (x, y) in the x-y plane is given by V = –Kxy. The field intensity at a distance r in this plane, from the origin is proportional to.

(1) r^2

(2) r

(3) 1/r

(4) $1/r^2$

4) A proton is released from rest, 10 cm from a charged sheet carrying charged density of -2.21×10^{-9} C/m^2. It will strike the sheet after the time (approximately) :

(1) 4μs

(2) 2μs

(3) $2\sqrt{2}\mu s$

(4) $4\sqrt{2}\mu s$

5) An infinite non-conducting sheet has a surface charge density $\sigma = 0.1\ \mu C/m^2$. How far apart are equipotential surfaces whose potentials differ by 50 V ?

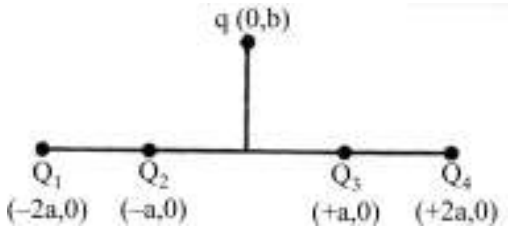
(1) 8.8 mm

(2) 8.8 cm

(3) 8.8 μm

(4) 8.8 pm

6) Four charges Q_1, Q_2, Q_3 and Q_4 of same magnitude are fixed along the x axis at x = – 2a, –a, +a and +2a, respectively. A positive charge q is placed on the positive y axis at a distance b > 0. Four options of the signs of these charges are given in List I. The direction of the forces on the charge q is given in List II. Match List I with List II and select the correct answer using the options given below the lists.



List I		List II	
P	Q_1, Q_2, Q_3, Q_4 all positive	a	+x
Q	Q_1, Q_2 positive; Q_3, Q_4 negative	b	–x
R	Q_1, Q_4 positive; Q_2, Q_3 negative	c	+y
S	Q_1, Q_3 positive; Q_2, Q_4 negative	d	–y

(1) P-c, Q-a, R-d, S-b

(2) P-d, Q-b, R-c, S-a

(3) P-c, Q-a, R-b, S-d

(4) P-d, Q-b, R-a, S-c

7) Let there be a spherically symmetric charge distribution with charge density varying as $\rho(r) = \rho_0 \left(\frac{5}{4} - \frac{r}{R}\right)$ upto r = R, and $\rho(r) = 0$ for r > R, where r is the distance from the origin. The electric potential at a distance R from the origin is given by :

(1) $\frac{\rho_0 r}{3\varepsilon_0}\left(\frac{5}{4} - \frac{r}{R}\right)$

(2) $\frac{4\pi\rho_0 r}{3\varepsilon_0}\left(\frac{5}{3} - \frac{r}{R}\right)$

(3) $\frac{\rho_0 R^2}{6\varepsilon_0}$

(4) $\frac{\rho_0 R^2}{3\varepsilon_0}$

8) Two isolated metallic solid spheres of radii R and 2R are charged such that both have same charge density σ. The spheres are then connected by a thin conducting wire. If the new charge density of the bigger sphere is σ'. The ratio $\frac{\sigma'}{\sigma}$ is :

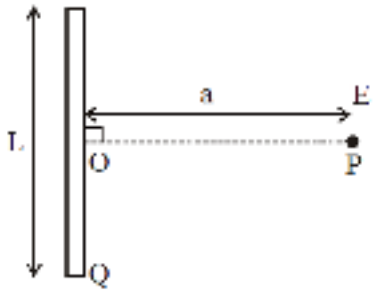
(1) $\frac{9}{4}$

(2) $\frac{4}{3}$

(3) $\frac{5}{3}$

(4) $\frac{5}{6}$

9) Find the electric field at point P (as shown in figure) on the perpendicular bisector of a uniformly charged thin wire of length L carrying a charge Q. The distance of the point P from the centre of the rod is $a = \frac{\sqrt{3}}{2}L$



(1) $\frac{\sqrt{3}Q}{4\pi\varepsilon_0 L^2}$

(2) $\frac{Q}{3\pi\varepsilon_0 L^2}$

(3) $\frac{Q}{2\sqrt{3}\pi\varepsilon_0 L^2}$

(4) $\frac{Q}{4\pi\varepsilon_0 L^2}$

10) An electric dipole is placed on x-axis in proximity to a line charge of linear charge density 3.0×10^{-6} C/m. Line charge is placed on z-axis and positive and negative charge of dipole is at a distance of 10 mm and 12 mm from the origin respectively. If total force of 4 N is exerted on the dipole, find out the amount of positive or negative charge of the dipole.

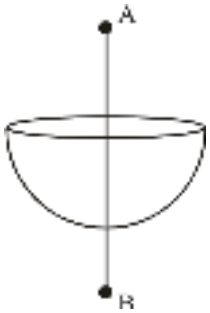
(1) 815.1 nC

(2) 8.8 μC

(3) 0.485 mC

(4) 4.44 μC

11) The diagram shows a uniformly charged hemisphere of radius R. It has volume charge density ρ. If the electric field at a point 2R distance above its center is E then what is the electric field at the



point which is 2R below its center?

(1) $(\rho R/6\varepsilon_0) + E$

(2) $(\rho R/12\varepsilon_0) - E$

(3) $(-\rho R/6\varepsilon_0) + E$

(4) $(\rho R/24e_0) + E$

12) Electric field due to a charged ring of radius R is maximum on the axis at a distance from circumference of the ring :

(1) $\frac{R}{\sqrt{2}}$

(2) $\sqrt{\frac{3}{2}}R$

(3) $\frac{\sqrt{3}}{2}R$

(4) $\sqrt{2}R$

13) A particle of mass 6.4×10^{-27} kg and charge 3.2×10^{-19} C is situated in a uniform electric field of 1.6×10^5 Vm^{-1}. The velocity of the particle at the end of 2×10^{-2} m path when it starts from rest is :

(1) $2\sqrt{3} \times 10^5$ ms^{-1}

(2) 8×10^5 ms^{-1}

(3) 16×10^5 ms^{-1}

(4) $4\sqrt{2} \times 10^5$ ms^{-1}

14) Potential difference between two concentric spherical shells is 10 V and charge in inner shell is 2μC. If 6μC charge is added in inner shell then what will be new potential difference between two shells :

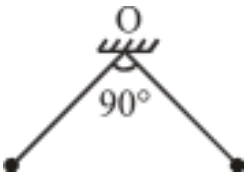
(1) 10 V

(2) 40 V

(3) 60 V

(4) zero

15) Two small spherical balls each carries a charge 10 μC are suspended by two insulting threads of equal length $\sqrt{2}$m each, from a fixed point O. In equilibrium threads are separated by an angle 90°

between them. Mass of each ball is :- 

(1) 0.1 gm

(2) 22.5 gm

(3) 1 gm

(4) 102.5 gm

16) A point charge 10μc is at a distance 5 cm directly above the centre of a square of side 10 cm. What is the magnitude of the electric flux through the squares :

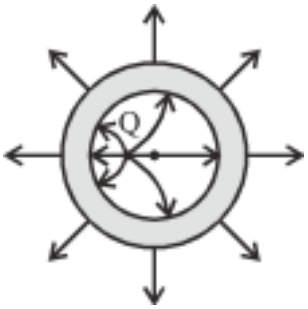
(1) $4.4 \times 10^6 \frac{Nm^2}{C}$

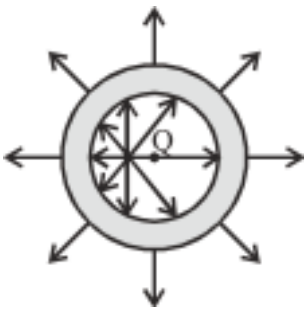
(2) $2.1 \times 10^8 \frac{Nm^2}{C}$

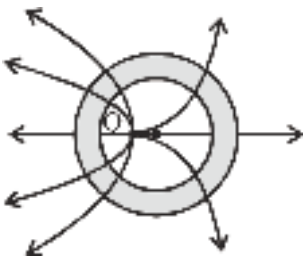
(3) $2.2 \times 10^{10} \frac{Nm^2}{C}$

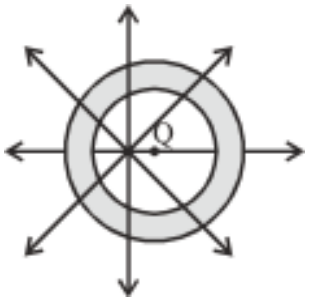
(4) $1.88 \times 10^5 \frac{Nm^2}{C}$

17) A point charge 'Q' is placed at a point inside a hollow conducting sphere. Which of the following electric field lines pattern is correct ?

(1) 

(2) 

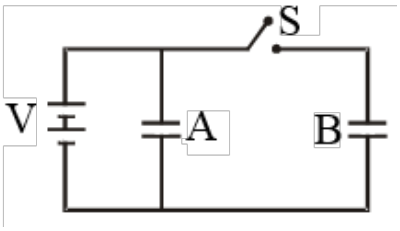
(3) 

(4) 

18) An electric charge 10^{-6}μC is placed at origin (0, 0) m of X-Y co-ordinate system. Two points P and Q are situated at $(\sqrt{3}, \sqrt{3})$m and $(\sqrt{6}, 0)$m respectively. The potential difference between the points P and Q will be :

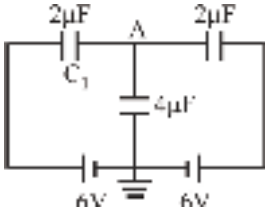
(1) $\sqrt{3}$V
(2) $\sqrt{6}$V
(3) 0 V
(4) 3 V

19) Figure given below shows two identical parallel plate capacitors connected to a battery with switch S closed. The switch is now opened and the free space between the plate of capacitors is filled with a dielectric of dielectric constant 3. What will be the ratio of total electrostatic energy stored in both capacitors before and after the introduction of the dielectric :-



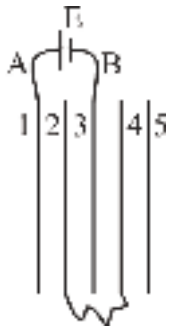
(1) 3 : 1
(2) 5 : 1
(3) 3 : 5
(4) 5 : 3

20) Three capacitors are connected as shown in fig. Then the charge on capacitor C_1 is :-



(1) 6μC
(2) 12μC
(3) 18μC
(4) 24μC

21) Five conducting parallel plates having area A and separation between them d, are placed as shown in the figure. Plate number 2 and 4 are connected by a wire and between point A and B, a cell



of emf E is connected. The charge flown through the cell is :-

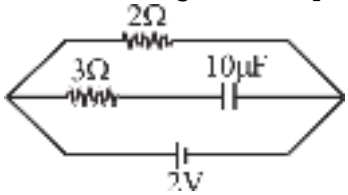
(1) $\frac{3}{4}\frac{\varepsilon_0 AE}{d}$

(2) $\frac{2}{3}\frac{\varepsilon_0 AE}{d}$

(3) $\frac{4\varepsilon_0 AE}{d}$

(4) $\frac{\varepsilon_0 AE}{2d}$

22) The charge on a capacitor of capacitance 10μF connected as shown in figure at steady state is :

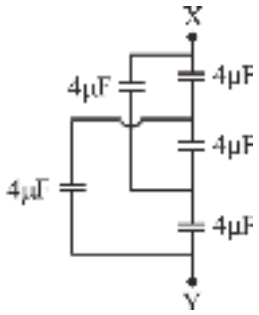


(1) 20 μC

(2) 15 μC

(3) 10 μC

(4) 0



23) Equivalent capacitance between X and Y points in the given figure is :-

(1) 2 μF

(2) 4 μF

(3) 3 μF

(4) 1 μF

24) ***Assertion (A) :-*** If the distance between parallel plates of a capacitor is halved and dielectric constant is made three times, then the capacitance becomes 6 times.
Reason (R) :- Capacity of the capacitor does not depend upon the nature of the material filled between the plates.

(1) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**

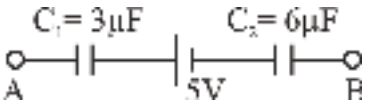
(2) **(A)** is true but **(R)** is false

(3) **(A)** is false but **(R)** is true

(4) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

25)

If $V_A - V_B = 7V$ then find stored energy in C_1



(1) $\frac{4}{3}\mu J$

(2) 6μJ

(3) $\frac{8}{3}\mu J$

(4) $\frac{10}{3}\mu J$

26) To form a combination of equivalent capacitance of 6 μF, with voltage of 1000 V, how many minimum number of capacitors, capacitance of each 3 μF and breakdown voltage of 400 V are required-

(1) 6

(2) 12

(3) 18

(4) 24

27) When a current of 2 A flows in a battery from negative to positive terminal, the potential difference across it is 12 V. If a current of 3 A flows in the opposite direction potential difference across the terminals of the battery is 15 V, the emf of the battery is

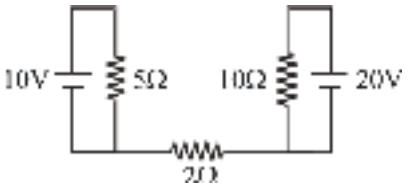
(1) 12.6 V

(2) 13.2 V

(3) 13.5 V

(4) 14.0 V

28) Find the current through 2Ω :-



(1) zero

(2) 1 amp

(3) 2 amp

(4) 4 amp

29) n identical bulbs are rated P watt and V volt. When they are connected in series, total power consumed is 20 watt and when connected in parallel they consume total power of 2000 watt. How many bulbs are used :-

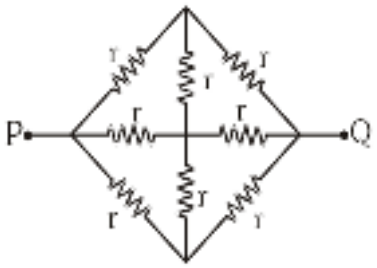
(1) 200 bulbs

(2) 100 bulbs

(3) 20 bulbs

(4) 10 bulbs

30) The equivalent resistance between the points P and Q in the network given here is equal to



(given $r = \frac{3}{2}\Omega$) :-

(1) $\frac{1}{2}\Omega$

(2) 1Ω

(3) $\frac{3}{2}\Omega$

(4) 2Ω

31) Current density $(\vec{J})$ at an area $\vec{A} = (\hat{i} + 4\hat{j})$ mm^2 is $\vec{J} = (2\hat{i} + 3\hat{j})$A/m^2. The current through the area is

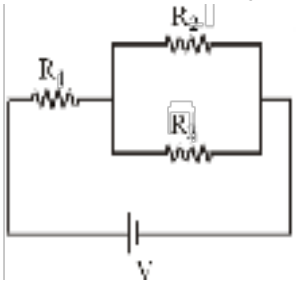
(1) 6 μA

(2) 4 μA

(3) 12 μA

(4) 14 μA

32) Three identical resistors $R_1 = R_2 = R_3$ are connected as shown to a battery of constant e.m.f. The power dissipated is :-



(1) The least in R_1

(2) Greatest in R_1

(3) In the ratio 1 : 2 in resistances R_1 and R_2 respectively

(4) The same in R_1 and in the parallel combination of R_2 and R_3

33) If in the experiment of wheat stone bridge, the position of the cell and galvanometer are interchanged then balance condition will :

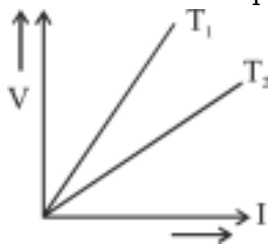
(1) Change

(2) Remains unchanged

(3) Depend on internal resistance of cell

(4) Depend on resistance of galvanometer

34) The voltage V and current I graph for a conductor at two different temperatures T_1 and T_2 are shown in the figure. The relation between T_1 and T_2 is :-



(1) $T_1 > T_2$

(2) $T_1 \approx T_2$

(3) $T_1 = T_2$

(4) $T_1 < T_2$

35) **Column-I** gives certain physical terms associated with flow of current through a metallic conductor.
Column-II gives some mathematical relations involving electrical quantities.
Match column-I and column-II with appropriate relations.

Column-I	Column-II
(A) Drift velocity	(P) $\frac{m}{ne^2\rho}$
(B) Electrical Resistivity	(Q) nev_d
(C) Relaxation Period	(R) $\frac{eE}{m}\tau$
(D) Current Density	(S) $\frac{E}{J}$

	A	B	C	D
(1)	R	Q	S	P
(2)	R	S	P	Q
(3)	R	S	Q	P
(4)	R	P	S	Q

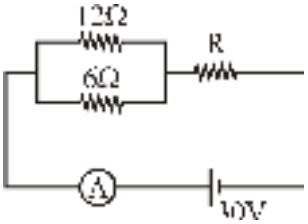
(1) 1

(2) 2

(3) 3

(4) 4

36) If the ammeter in the given circuit reads 5A, the resistance R is :-



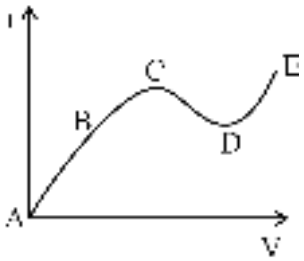
(1) 1Ω

(2) 4Ω

(3) 5Ω

(4) 2Ω

37) From the graph between current i & voltage V shown, identify the portion corresponding to negative resistance :



(1) DE

(2) CD

(3) BC

(4) AB

38) Two unknown resistances X and Y are connected to left and right gaps of a meter-bridge and the balancing point is obtained at 80 cm from left. When a 10Ω resistance is connected in parallel to X, the balancing point is 50 cm from left, the values of X and Y respectively are :-

(1) 40Ω , 9Ω

(2) 30Ω , 7.5Ω

(3) 20Ω , 6Ω

(4) 10Ω , 3Ω

39) An electric kettle has two heating coils. When one of the two coils is switched on, the tea begins to boil in 10 minutes. When second coil only is switched on, the boiling begins in 15 minutes. When both coils are connected in parallel and switched on the tea starts boiling in time t, then t is :-

(1) 4 minute

(2) 6 minute

(3) 25 minute

(4) 12 minute

40) A wire of resistance R is divided in 10 equal parts. These parts are connected in parallel, the equivalent resistance of such connection will be :-

(1) 0.01 R

(2) 0.1 R

(3) 10 R

(4) 100 R



41) The potential difference between A and B as shown in figure is :-

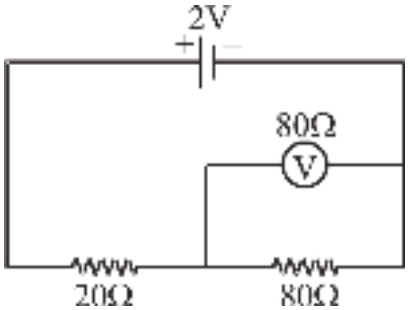
(1) 1 V

(2) 2 V

(3) 3 V

(4) 4 V

42) In the adjoining figure the emf of the cell is 2 V and internal resistance is negligible. The resistance of the voltmeter is 80 ohm. The reading of voltmeter will be :-



(1) 2.00 volt

(2) 1.33 volt

(3) 1.66 volt

(4) 0.80 volt

43) A battery of 10 V and internal resistance 0.5 Ω is connected across a variable resistance R. The value of R for which the power delivered become maximum, is equal to :-

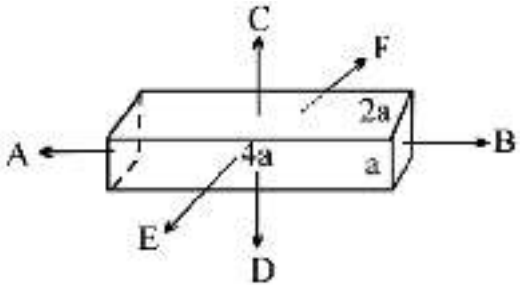
(1) 0.5 Ω

(2) 1 Ω

(3) 1.5 Ω

(4) 2 Ω

44) A conductor with rectangular cross section has dimensions (a × 2a × 4a) as shown in figure. Resistance across AB is x, across CD is y and across EF is z. Then



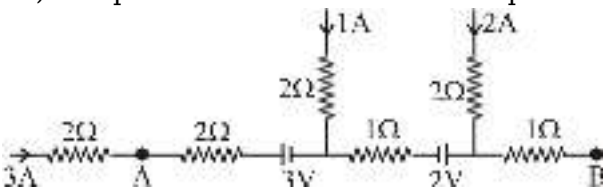
(1) x = y = z

(2) x > y > z

(3) y > x > z

(4) x > z > y

45) The potential difference between points A and B, in a section of a circuit shown, is



(1) 5 volt

(2) 1 volt

(3) 10 volt

(4) 17 volt

CHEMISTRY

1) Consider aqueous solution of 0.2M glucose, 0.1M Na_3PO_4 and 0.05M $Al_2(SO_4)_3$. What is correct about these solution? (Assume m = M)
(a) The vapour pressure and freezing point are the lowest for glucose.
(b) The vapour pressure and freezing point are the highest for glucose.
(c) The elevation in boiling point is the highest for $Al_2(SO_4)_3$.
(d) The depression in freezing point is the highest for Na_3PO_4.

(1) a, b, c

(2) b, c

(3) only a

(4) b & d

2) At 298 K given specific conductance of saturated aq. Ag_2CrO_4 solution is $1 \times 10^{-6}\ \Omega^{-1}cm^{-1}$. If equivalent conductance of saturated Ag_2CrO_4 solution 100 $\Omega^{-1}cm^2$ equivalent^{-1}. Thus solubility product K_{sp} of Ag_2CrO_4.

(1) 4×10^{-5}

(2) 5×10^{-4}

(3) 5×10^{-16}

(4) 15×10^{-4}

3) The metal that cannot be obtained by electrolysis of an aqueous solution of its salt is :

(1) Cu
(2) Cr
(3) Ca
(4) Ag

4) One molal solution of a carboxylic acid in benzene shows the elevation of boiling point of 1.5 K. The degree of association for dimerisation of the acid in benzene is (K_b for benzene = 2.5 K kg mol^{-1}) :-

(1) 60%
(2) 70%
(3) 75%
(4) 80%

5) Which of the following is correct statement ?
(A) Edema is puffiness or swelling caused due to people taking a lot of salt.
(B) 0.9% mass/volume solution of NaCl is isotonic to human blood.
(C) Preservation of meat can be done by salting.
(D) Osmotic pressure is best method to determine molecular weight of biomolecule because they are stable at high temperature and polymers have high solubility.

(1) A, C, D
(2) B, C, D
(3) B, C
(4) A, B, C

6) Resistance of 0.2 M solution of an electrolyte is 80Ω. The specific conductance of the solution is 1.4 Sm^{-1}. If resistance of the 0.4 M solution of the same electrolyte is 280Ω. It's molar conductivity is Sm^2 mol^{-1} is :-

(1) 10
(2) 10^{-3}
(3) 10×10^{-2}
(4) 10^{-4}

7) For elementary reaction A → 2B it takes 35 minute for the concentration of A to drop from 2.5 M to 1.5 M. If initial concentration of A is 8 M then time taken for concentration of B to increase from zero to 6.4 M [Volume of container 1 L in both case] :-

(1) 70 minute
(2) 105 minute
(3) 35 minute
(4) 140 minute

8) Henry's law constant for CO_2 in water is $5/3 \times 10^8$ Pa at 298 K. Calculate the quantity of CO_2 in

1500 ml of soda water when packed under 2.5 atm CO_2 pressure at 298 K.

(1) 5.56×10^{-5} g

(2) 5.56 g

(3) 2.28×10^{-5} g

(4) 0.125 g

9) If 24 g NaOH is dissolved in 372 g H_2O and this solution is freezed to –4°C. Calculate mass of H_2O separated as ice [K_f = 1.86, K/Molality]

(1) 72 g

(2) 36 g

(3) 200 g

(4) No ice will be separated

10) An increase in equivalent conductance of $K_4[Fe(CN)_6]$ with dilution is mainly due to (Assume 100 % ionisation)-

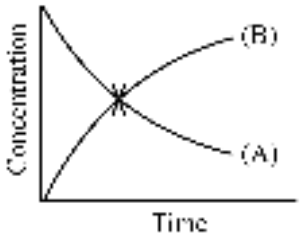
(1) increase in number of ions

(2) increase in ionic mobility of ions

(3) Increase in resistance

(4) increase in both, number of ions and ionic mobility of ions.

11) Figure depict the change in concentration of species A & B for reaction 2A → 3B, as a function of time the point of intersection of two curve represent ?



(1) $t_{20\%}$

(2) $t_{40\%}$

(3) $t_{60\%}$

(4) $t_{80\%}$

12) Equal weight of a same solute are dissolved in equal weight of two solvent A & B and formed very dilute solution. The relative lowering of vapour pressure for the solution B is thrice the relative lowering of vapour pressure for the solution A. If M_A and M_B are the molecular weight of solvent A & B respectively then :-

(1) $M_A = M_B$

(2) $M_B = 3M_A$

(3) $M_A = 4M_B$

(4) $M_A = 3M_B$

13) On increasing temperature from 300 K to 340 K, rate of reaction A increase by 6 times and rate of reaction B increase by 216 times then correct relationship between activation energy of A & B is :-

(1) $E_A = 3E_B$

(2) $E_B = 2E_A$

(3) $3E_A = E_B$

(4) $E_A = 2E_B$

14) Which of the following are correctly matched ?

	Electrolyte	**Product at anode**	**Product at cathode**
(1)	Aqueous NaCl	O_2	H_2
(2)	Molten $MgCl_2$	Cl_2	H_2
(3)	Dilute NaCl	O_2	H_2
(4)	Aqueous $AgNO_3$	O_2	H_2

(1) 1

(2) 2

(3) 3

(4) 4

15) For a homogeneous gaseous reaction A → 3B, if pressure after time t was P_t and after completion of reaction, pressure was P_∞, then select correct relation.

(1) $K = \frac{1}{t}\ell n\left(\frac{P_\infty}{3(P_\infty - P_t)}\right)$

(2) $K = \frac{1}{t}\ell n\left(\frac{2P_\infty}{P_\infty - P_t}\right)$

(3) $K = \frac{1}{t}\ell n\left(\frac{3P_\infty}{2P_\infty - P_t}\right)$

(4) $K = \frac{1}{t}\ell n\left(\frac{2P_\infty}{3(P_\infty - P_t)}\right)$

16) At 25°C the vapour pressure of pure liquid A (mol. wt. = 40) is 100 torr while that of pure liquid B is 40 torr (mol. wt. = 80). If a solution is made by mixing 20g of each A and B then mole fraction of B in vapour phase is-

(1) $\frac{3}{4}$

(2) $\frac{5}{6}$

(3) $\frac{2}{3}$

(4) $\frac{1}{6}$

17) Total vapour pressure of mixture of 1 mol X (P^o_x = 150 torr) and 2 mol Y (P^o_y = 300 torr) is 240 torr. Which is/are true for this solution.
(A) There is negative deviation from Raoult's law.
(B) There is positive deviation from Raoult's law.
(C) Heat is released in the formation of solution.
(D) Azeotrope of this mixture has boiling point greater than X and Y.

(1) B, C
(2) Only B
(3) A, B, C
(4) A, C, D

18) **Assertion :** The order of a reaction can have fractional value.
Reason : The order of a reaction is given by a balanced equation

(1) Both assertion and reason are true and reason is a correct explanation of assertion
(2) Both assertion and reason are true but reason is not a correct explanation of assertion
(3) Assertion is true but reason is false
(4) Assertion is false but reason is true

19) The SRP of Cu^{2+}/Cu is +0.34V. Calculate reduction potential at pH = 12 for the above couple in $Cu(OH)_2$ solution. K_{sp} of $Cu(OH)_2$ is 1×10^{-19}.

(1) –0.45 V
(2) 0.45 V
(3) –0.11 V
(4) +0.11 V

20) 40 ml of H_2O_2 is sufficient for reaction with 20 ml of 0.2M $KMnO_4$ in acidic medium. What is volume strength of H_2O_2 ?

(1) 1.4 V
(2) 11.2 V
(3) 5.6 V
(4) 2.8 V

21) The reaction (A) $\xrightarrow{k}$ products, is zero order while the reaction (B) $\xrightarrow{k}$ products, is first order if initial concentration of A is 0.693 M then :-

(1) $(t_{1/2})_A = (t_{1/2})_B$
(2) $2(t_{1/2})_A = (t_{1/2})_B$
(3) $(t_{1/2})_A = 2(t_{1/2})_B$
(4) $4(t_{1/2})_A = (t_{1/2})_B$

22) Given

$$E^o_{Au^{3+}/Au} = +1.40V$$

$E^{o}_{Cu^{2+}/Cu} = +0.34V$

$E^{o}_{Zn^{2+}/Zn} = -0.76V$

$E^{o}_{Ni^{2+}/Ni} = -0.25V$

Which of the following statement is incorrect?

(1) $AuCl_3$ cannot be store in copper vessel.

(2) $ZnSO_4$ can be store in Nickle vessel.

(3) $Ni(NO_3)_2$ can be store in Zn vessel.

(4) $CuSO_4$ cannot be store in Nickle vessel.

23) The concentration of A in the reaction A → B was measured as a function of time and the following data is obtained

[A] molar	8	6	4	3
t min	0.0	0.05	0.1	0.125

The order of reaction is

(1) 3

(2) 2

(3) 0

(4) 1

24) Which have highest osmotic pressure?

(1) 1M NaCl

(2) 1N K_2SO_4

(3) 1N Na_3PO_4

(4) 1M K_2SO_4

25) An aqueous solution of glucose containing 36 g of glucose in 1836 g of solution. Then molality of solution is -

(1) 0.111

(2) 0.222

(3) 0.333

(4) 0.444

26) For a first order reaction A → P, the temperature (T) and dependent rate constant (K), was found to follow the equation $\log K = -(1500)\frac{1}{T} + 7$. The pre-exponential factor A and the activation energy E_a, respectively are

(1) 1×10^6, 28.72 kJ/mol

(2) 28.72 kJ/mol, 1×10^7

(3) 10^7, 28.7 kJ/mol

(4) 10^6, 287 kJ/mol

27) For the following half cell reaction E° values are

$Mn^{2+}_{(aq)} + 2H_2O_{(\ell)} \rightarrow MnO_{2(s)} + 4H^+ + 2e^-$

E° = –1.23 V

$MnO^-_4(aq.) + 4H^+_{(aq)} + 3e^- \rightarrow MnO_{2(s)} + 2H_2O_{(\ell)}$

E° = +1.70 V

Thus which statement is correct ?

(1) Mn^{2+} react with MnO_2 to form MnO^-_4

(2) Mn^{2+} react with MnO^-_4 in acid solution to form MnO_2

(3) $Mn(MnO_4)_2$ is stable in acid solution

(4) MnO_2 disproportionate to Mn^{2+} and MnO^-_4 in acid solution

28) For a reaction of iodide ion with H_2O_2 at room temperature, what will be the overall order of reaction?

(1) First order

(2) Second order

(3) Third order

(4) Zero order

29) What is colour change at the end point in iodometric titration?

(1) Blue colour appear

(2) Dark green colour appear

(3) Blue colour disappear

(4) Green colour disappear

30) **Assertion :** In the titration between HCl (burette) and NaOH (conical flask) if we use phenolpthalein as indicator then pink colour at end point appear.
Reason : At end point unionised part of indicator dominant in above case.

(1) Assertion and Reason both correct and Reason correctly explain Assertion.

(2) Assertion is correct but Reason is wrong.

(3) Assertion is wrong but Reason is correct.

(4) Both Assertion and Reason wrong.

31) What set is correctly match ?

I		II	
(A)	As_2S_3 sol	(P)	Strong affinity with solvent
(B)	Starch sol	(Q)	Reversible sol

(C)	Gelatin sol	(R)	Special methods required for preparation
(D)	$Fe(OH)_3$ sol	(S)	Lyophobic sol

(1) (A)-R, (B)-P, (C)-Q, (D)-S

(2) (A)-P, (B)-Q, (C)-R, (D)-S

(3) (A)-S, (B)-P, (C)-R, (D)-Q

(4) (A)-R, S, (B)-P, Q, (C)-P, Q, (D)-R, S

32) The density of 3 M solution of sodium thiosulphate ($Na_2S_2O_3$) is 1.58 g/ml. Calculate the amount of $Na_2S_2O_3$ by w/w % :

(1) 20%

(2) 40%

(3) 55%

(4) 30%

33) The reaction A → Product, follow first order kinetics, in 21 min the concentration of A changes from 0.8 M to 0.1 M. The rate of reaction (in M min^{-1}) when concentration of A is 0.05 M :-

(1) 2×10^{-3}

(2) 4.95×10^{-3}

(3) 1×10^{-3}

(4) 10×10^{-3}

34) $E^{o}_{Cr^{3+}/Cr} = -0.74\ V$, $E^{o}_{MnO_4^{\ominus}/Mn^{2+}} = 1.51\ V$
$E^{o}_{Cr_2O_7^{2-}/Cr^{3+}} = 1.33\ V$, $E^{o}_{Cl_2/Cl^{\ominus}} = 1.36\ V$
According to above data which is incorrect :

(1) Oxidising power $MnO_4^{\ominus} > Cl_2$

(2) Reducing power $Cr > Mn^{2+}$

(3) Oxidising power $Cr_2O_7^{2-} < MnO_4^{\ominus}$

(4) Reducing Power $Cl^{\ominus} > Cr^{3+}$

35) If 50 ml burette is filled with 0.5 M $K_2Cr_2O_7$ which is used for titration with 25 ml, 0.3 M ferrous ammonium sulphate $(NH_4)_2Fe(SO_4)_2$. What would be volume of $K_2Cr_2O_7$ remain in burette at end point :-

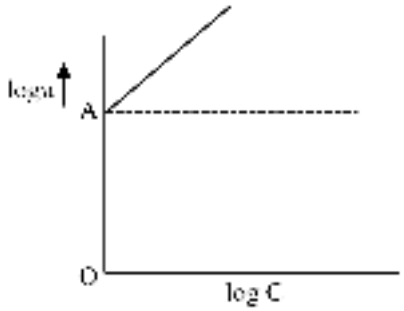
(1) 2.5 ml

(2) 47.5 ml

(3) 42.5 ml

(4) 7.5 ml

36) Variation of log π (atm) v/s log C (concentration of solute in g/L) is given following :



At 300 K, for a solute of molar mass 150 g/mol OA = log(x × 10^{-2}).
What is the value of x :

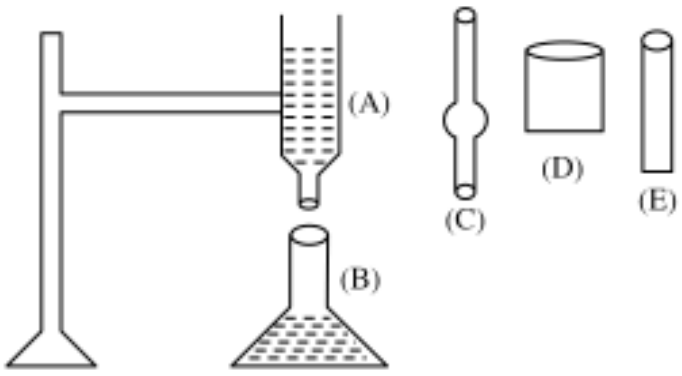
(1) 4.1
(2) 8.21
(3) 1.64
(4) 16.42

37) $N_2 + 3H_2 \rightarrow 2NH_3$ if rate of formation of NH_3 is 6.8 kg min^{-1} hence rate at which H_2 react is :-

(1) 10.2 kg/min
(2) 1.2 kg/min
(3) 3.4 kg/min
(4) 0.6 kg/min

38) $2C^{+}_{(aq)} + A_{(s)} \rightleftharpoons 2C_{(s)} + A^{+2}_{(aq)}$
If K_{eq} of reaction is $10^{12.7}$ तथा $E^{o}_{A^{2+}/A}$ = 0.75 V. Then calculate E°(C^{+}/C) will be :-

(1) 1.125 V
(2) 0.35 V
(3) 0.375 V
(4) 0.65

39) 

Which is incorrect statement :

(1) B is conical flask and C is pipette
(2) A is burette and D is beaker
(3) A is pipette and C is burette
(4) E is test tube and B is conical flask

40) Find concentration of 20 ml $Ca(OH)_2$ which is require for equivalence point with 40 ml of 0.5 M HCl :

(1) 0.25 M
(2) 1 N
(3) 0.5 N
(4) 1 M

41) Vapour pressure of CCl_4 and CH_2Cl_2 at 25°C are 150 mm Hg and 420 mm Hg respectively. Vapour pressure of the solution obtained by mixing 77 g of CCl_4 and 85 g of CH_2Cl_2 at the same temperature will be (mol. wt CCl_4 = 154 u, mol wt. CH_2Cl_2 = 85 u) :

(1) 180.7 mm Hg
(2) 120.6 mm Hg
(3) 330 mm Hg
(4) 612.3 mm Hg

42) Which of the following statement is correct for reaction A + 3B → 2C :

(1) Rate of disappearance of B is twice the rate of disappearance of A.
(2) Rate of appearance of C is twice the rate of disappearance of B.
(3) Rate of reaction is thrice of rate of disappearance of B.
(4) Rate of reaction is half of the rate of appearance of C.

43) If $E^{o}_{A^{2+}/A}$ = –0.54 V and
$E^{o}_{A^{3+}/A^{2+}}$ = +0.84 V
The standard EMF of the reaction
$A + 2A^{3+} \rightarrow 3A^{2+}$ will be :-

(1) 2.56 V
(2) 1.38 V
(3) 5.42 V
(4) 0.56 V

44) The heat of solution of anhydrous $CuSO_4$ and Enthalpy of hydration of anhydrous $CuSO_4$ are –66.74 kJ/mol and –78.50 kJ/mol respectively. Then what is enthalpy of solution of hydrated $CuSO_4$:

(1) –18.69 Kcal/mol
(2) 11.76 Kcal/mol
(3) –11.76 Kcal/mol
(4) +18.69 Kcal/mol

45) Which will form +ve colloid of lyophobic AgI ?

(1) 50 ml, 0.1 M $AgNO_3$ + 50 ml, 0.1 M KI
(2) 75 ml, 0.1 M $AgNO_3$ + 50 ml, 0.1 M KI

(3) 40 ml, 0.1 M $AgNO_3$ + 20 ml, 0.2 M KI

(4) 50 ml, 0.1 M $AgNO_3$ + 50 ml, 0.2 M KI

BIOLOGY

1) In which of the following plant, both Autogamy and Geitonogamy are prevented ?

(1) Date palm

(2) Castor

(3) Coconut

(4) Maize

2) In a flowering plant, the ___(A)___ represents the male gametophytic generation whereas ___(B)___ represents the female gametophytic generation. A & B are respectively -

(1) Pollen grain & Ovule

(2) Pollensac & Nucellus

(3) Pollen grain & Embryosac

(4) Pollensac & Embryosac

3) The pollen tube releases both male gametes inside the cytoplasm of which degenerating cell of embryo sac ?

(1) Antipodals cell

(2) Synergid cell

(3) Egg cell

(4) Central cell

4) In *Zostera* pollination takes place through _____

(1) Wind

(2) Water

(3) Animal

(4) Insect

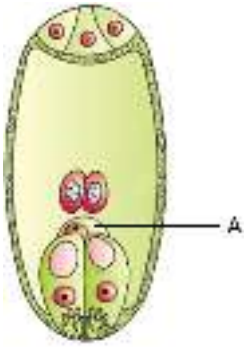
5) _____ pollinated flowers often have a single ovule in each ovary and numerous flowers are packed into an inflorescence.

(1) Wind

(2) Water

(3) Insect

(4) Bird



6) In which of the following option labelled part 'A' is correctly identified?

(1) Egg cell
(2) Embryo
(3) Synergids cells
(4) Zygote

7) Which of the following event is not included under post fertilization events in flowering plants ?

(1) Development of embryo
(2) Development of endosperm
(3) Transformation of ovary into fruit
(4) Formation of embryosac

8) Select the incorrect statement regarding angiosperm plants.

(1) Megaspore is diploid
(2) Megaspore is the first cell of female gametophytic generation
(3) Microspore is the first cell of male gametophytic generation
(4) Development of seed is a post fertilization event

9) Read the following statement (A to D)
(A) Pollen grains are generally 25-50 micrometers in diameter
(B) The megasporangium develop further and become pollen sac.
(C) In rice and wheat pollen grains lose viability within 30 seconds of their release
(D) The body of the ovule fuses with funicle in the region called hilum.
Which of the above statements are incorrect ?

(1) A and B
(2) B and C
(3) A and D
(4) B, C and D

10) Which of the following statement is incorrect about the flowering plant ?

(1) Self incompatibility is a device to prevent inbreeding
(2) The central cell of embryo sac after triple fusion becomes the primary endosperm cell
(3) In dicotyledonous embryo, the portion of embyonal axis below the level of cotyledons is the epicotyle

(4) Pollen grains can be stored at –196°C

11) **Statement-I :-** Monoecious plant with unisexual flower prevents autogamy but not geitonogamy.
Statement-II :- Dioecious plant prevents both autogamy and geitonogamy.

(1) Both statement-I and statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but statement-II is correct.
(4) Both statement-I and statement-II are correct.

12) **Statement-I :-** Formation of embryosac from a single megaspore is termed as monosporic development.
Statement-II :- In majority of flowering plants, one of the megaspore is functional while other three degenerate.

(1) Both statement-I and statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but statement-II is correct.
(4) Both statement-I and statement-II are correct.

13) **Assertion (A) :** Geitonogamy from the genetical point of view is similar to autogamy.
Reason (R) : In Geitonogamy pollen grains come from the same plant.
Choose the correct option:-

(1) Both **A** and **R** are true but **R** is not the correct explanation of **A**.
(2) **A** is true but **R** is false.
(3) **A** is false but **R** is true.
(4) Both **A** and **R** are true but **R** is the correct explanation of **A**.

14) Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion (A) : The generative cell is small and floats in the cytoplasm of the vegetative cell.
Reason (R) : Cells of the nucellus have abundant reserve food materials.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(3) **Assertion** is true but **Reason** is false.
(4) **Assertion** is false but **Reason** is true.

15) Match Column-I with Column-II and select the correct answer.

Column-I		**Column-II**	
A.	Sporopollenin	**I.**	The body of ovule fuses with funicle

B.	Egg apparatus	**II.**	Towards micropylar end of embryo sac
C.	Hilum	**III.**	Inner wall of microsporangium
D.	Tapetum	**IV.**	One of the most resistant organic material

(1) A-IV, B-I, C-II, D-III

(2) A-IV, B-II, C-I, D-III

(3) A-IV, B-III, C-II, D-I

(4) A-II, B-III, C-IV, D-I

16) Match the column-I with column-II

Column-I		**Column-II**	
(a)	Persistent Nucellus	(I)	Onion
(b)	Maize cotyledon	(II)	Pea
(c)	Endospermic seed	(III)	Scutellum
(d)	non-Endospermic seed	(IV)	Beet

(1) a-III,b-II,C-I,d-IV

(2) a-IV,b-III,C-I,d-II

(3) a-II,b-III,C-IV,d-I

(4) a-I,b-II,C-III,d-IV

17) Match the column-I with column-II and select the correct answer :

Column-I		**Column-II**	
(A)	Nucellus	(i)	Fascinating arrays of patern & design
(B)	Chalaza	(ii)	Soft & elastic in nature
(C)	Exine	(iii)	Basal part of the ovule
(D)	Intine	(iv)	Main part of ovule

	A	**B**	**C**	**D**
(1)	(iii)	(ii)	(i)	(iv)
(2)	(iv)	(i)	(ii)	(iii)
(3)	(iv)	(iii)	(i)	(ii)

(4)	(iii)	(iv)	(ii)	(i)

(1) 1

(2) 2

(3) 3

(4) 4

18) Which of the following statements are not true for the seeds?
A. Seeds can be easily stored for a longer period of time due to dormancy and dehydration.
B. Seeds form the basis of agriculture.
C. Seeds are the final product of asexual reproduction.
D. Formation of seeds in angiosperms is more dependable for continuity of species.
E. Seeds are usually genetically identical.

(1) A, C and D

(2) C and E

(3) A, B and D

(4) A, C, D and E

19) Some structures are given below which are related to life cycle of a typical angiosperms. In which how many structures are diploid ?
Endosperm, Nucellus, integument, pericarp, ovary, Zygote, Radicle, plumule, cotyledon, Egg cell, male gamete and root cap.

(1) Four

(2) Nine

(3) Seven

(4) Two

20) Arrange the following events in correct sequence for the development of male gametophyte in angiosperms :
(A) Formation of microspore mother cells.
(B) Meiosis to form microspore tetrads.
(C) Pollen grain with two male gametes.
(D) Pollen grain with generative and vegetative cells.
(E) Formation of microsporangium
Options :

(1) E → B → A → D → C

(2) E → A → B → D → C

(3) A → B → C → D → E

(4) D → C → E → B → A

21) Identify the set of correct statements :-
(A) On the basis of presence or absence of endosperm in mature seed it is of two types albuminous and non-albuminous seeds.
(B) The residual, persistent nucellus in the seed is called perisperm.
(C) As the seed matures its water content is reduced.

(D) A seed consists of seed coat, cotyledons and an embryonal axis.

(1) (A) and (C)
(2) (B) and (C)
(3) (A), (B) and (C)
(4) (A), (B), (C) and (D)

22) In dorsiventral leaves, the location of xylem and spongy tissue are towards :-

(1) Abaxial epidermis
(2) Adaxial Epidermis
(3) Adaxial and abaxial side respectively
(4) Abaxial and adaxial side respectively

23) Which of the following forms hypodermis of monocot stem ?

(1) Chlorenchyma
(2) Collenchyma
(3) Sclerenchyma
(4) Parenchyma

24) A dicot root differs from a monocot root in which of the following ?

(1) Presence of hypodermis
(2) Presence of epidermis
(3) Presence of less-developed (poorly developed) pith
(4) Presence of radial vascular bundle

25) Fill in the blanks and select the correct option :-
The root hairs are __(a)__ elongations of the root epidermal cells and the trichomes in the shoot system are usually __(b)__ .

(1) (a) multicellular, (b) unicellular
(2) (a) unicellular, (b) multicellular
(3) (a) unicellular, (b) unicellular
(4) (a) multicellular, (b) multicellular

26) **Statement-I :** Polyarch xylem bundles occurs in monocot root.
Statement-II : Large & well developed pith is found in dicot root.

(1) Both **Statement I** and **Statement II** are correct.
(2) Both **Statement I** and **Statement II** are incorrect.
(3) **Statement I** is correct but **Statement II** is incorrect
(4) **Statement I** is incorrect but **Statement II** is correct

27) Presence of water containing cavities within the vascular bundles is the characteristic of :-

(1) Monocot stem

(2) Dicot stem

(3) Dicot root

(4) Monocot leaf

28) In grasses, the guard cells are :-

(1) Barrel shaped

(2) Kidney shaped

(3) Oval shaped

(4) Dumb-bell shaped

29) **Statement-I :** .Vascular bundles of monocot stem are surrounded by scleranchymatous bundle sheath.
Statement-II : Vascular bundles of leaves are surrounded by parenchymatous bundle sheath.

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Statement-I is incorrect but statement-II is correct

30) Match the Column-I with Column-II & select correct option -

Column-I		**Column-II**	
(A)	Epidermal tissue system	(i)	Open vascular bundles, Radial vascular bundles, Close vascular bundles
(B)	Ground tissue system	(ii)	Trichome, Guard cells, Subsidiary cells
(C)	Vascular tissue system	(iii)	Cortex, Mesophyll, Pericycle

(1) (A–ii), (B–i), (C–iii)

(2) (A–ii), (B–iii), (C–i)

(3) (A–i), (B–ii), (C–iii)

(4) (A–iii), (B–ii), (C–i)

31) In a monocot stem :
I. Water containing cavities are present.
II. Vascular bundle conjoint and closed.
III. Phloem parenchyma absent.
IV. Hypodermis collenchymatous.

(1) Only I and II are correct

(2) Only II and III are correct

(3) I, II and III are correct

(4) II and IV are correct

32) Stony endocarp and shield shaped cotyledon are found respectively in -

(1) Maize and Coconut
(2) Mango and Maize
(3) Tomato and Gram
(4) Mango and Coconut

33) Which of the following pair of plant contain swollen placenta with superior oblique ovary?

(1) Petunia, Mustard
(2) Belladona, Pea
(3) Potato, Tomato
(4) Tobacco, Lily

34) In which of the following plant have hypogynous flower with parietal placentation.

(1) Plum
(2) Apple
(3) Mustard
(4) Cucumber

35) Match the list I with List-II

List-I		**List-II**	
A	Marginal placentation	I	*Calotropis*
B	Valvate aestivation	II	Bean
C	Zygomorphic	III	Lily
D	Trimerous flower	IV	*Cassia*

(1) A - I, B - II, C - IV, D - III
(2) A - II, B - III, C - I, D - IV
(3) A - I, B - II, C - III, D - IV
(4) A - II, B - I, C - IV, D - III

36) Monoadelphous condition of stamen and actinomorphic symmetry of flower is found in :-

(1) Brinjal
(2) *Citrus*
(3) Pea
(4) China rose

37) If gynoecium is situated in the centre and other parts of the flower are located on the rim of the thalamus almost at the same level, it is called :

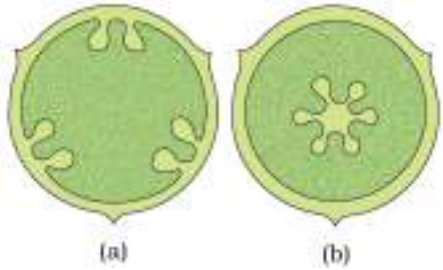
(1) Epigynous

(2) Perigynous

(3) Either perigynous or epigynous

(4) Neither perigynous nor epigynous

38) Identify the type of placentation based on arrangement of ovules within the ovary from the given figures (a) and (b)



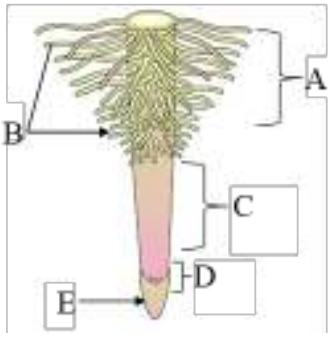
(1) (a) Marginal (b) Axile

(2) (a) Parietal (b) Free central

(3) (a) Axile (b) Basal

(4) (a) Basal (b) Marginal

39) What is the function of labelled structures region 'B' in the given diagram ?



(1) Storage of food.

(2) Absorption of minerals and water from soil.

(3) Photosynthesis .

(4) Preventing water loss due to transpiration.

40) **Assertion :-** Ovary is one-chambered but later on it becomes two-chambered in mustard.
Reason :- False septum is formed from thalamus in mustard.

(1) Both Assertion and Reason are correct and reason is the correct explanation of Assertion.

(2) Both assertion and reason are true but reason is not the correct explanation of assertion.

(3) Assertion is correct but reason is incorrect

(4) Assertion is incorrect but reason is correct

41) **Assertion :-** Twisted aestivation is found in chinarose and cotton.
Reason :- In chinarose and cotton, margins of sepals or petals overlap one another but not in any particular direction.

(1) Both Assertion and reason are true and reason is the correct explaination of assertion.

(2) Both assertion and reason are true but reason is not the correct explaination of assertion

(3) Assertion is true but reason is false.

(4) Assertion is false but reason is true.

42) Given below are the two statements :
Statement-I : In monocot plants primary root is short lived.
Statement-II : In monocot like wheat, fibrous root system is present.
Choose the correct option :

(1) Both the statements are correct

(2) Only statement -I is correct

(3) Only statement -II is correct

(4) Both the statements are incorrect

43) Read the following statements carefully :-
(I) A carpel consists of three parts namely stigma, style and ovary.
(II) Ovary is the enlarged basal part of carpel.
(III) Each ovary bears one or more ovules attached to a flattened, cushion like placenta.
(IV) When the ovules are borne on central axis and septa are absent the placentation is called axile.
(V) In *dianthus* and *primrose* ovary is one chambered but it becomes two chambered due to formation of false septum.
How many of the given statements are correct ?

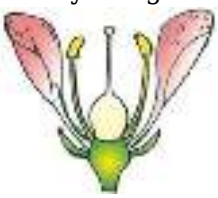
(1) 2

(2) 4

(3) 3

(4) 5

44) Identify the given figure below : (A)  (B)  (C) 

(D) 

(1) A-Racemose inflorescence,
B-Alternate phyllotaxy,
C-Fibrous root, D-Hypogynous flower

(2) A-Cymose inflorescence,
B-Alternate phyllotaxy,
C-Fibrous root, D-Hypogynous flower

(3) A-Cymose inflorescence,
B-opposite phyllotaxy,
C-Tap root, D-Epigynous flower

(4) A-Cymose inflorescence,
B-Opposite phyllotaxy,
C-Fibrous root, D-Perigynous flower

45) When stamens are attached to the petals they are called ___A___ and stamens are attached to the perianth they are called ___B___ and stamens are free they are called ___C___.
Select correct option for A,B and C and fill in the blanks:-

(1) A → Epiphyllous, B → Epipetalous,
C → diadelphous

(2) A → Polyandrous , B → Epiphyllous,
C → Epipetalous

(3) A → Monoadelphous, B → Polygandrous,
C → Epipetalous

(4) A → Epipetalous, B → Epiphyllous,
C → Polyandrous

46) **Assertion (A)** : LH surge is essential for ovulation in human females.
Reason (R): LH surge induces rupture of Graafian follicle and release of secondary oocyte.

(1) Both A and R are true, and R is the correct explanation of A

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, but R is false

(4) A is false, but R is true

47) Parturition is induced by :-

(1) Complex neural mechanism only

(2) Complex endocrine mechanism only

(3) Complex neuroendocrine mechanism

(4) Simple neuroendocrine mechanism

48) The first movements of the foetus and appearance of hair on the head are usually observed during the :-

(1) Third month

(2) Fourth month

(3) Fifth month

(4) Sixth month

49) Arrange the events of fertilisation in humans in the correct sequence.
(A) Syngamy
(B) Karyogamy
(C) Plasmogamy
(D) Completion of Meiosis-II

(E) Amphimixis
Choose the correct answer from the options given below :

(1) (C), (A), (E), (B), (D)

(2) (A), (D), (E), (C), (B)

(3) (A), (D), (C), (B), (E)

(4) (B), (A), (E), (D), (C)

50) Which of the following event(s) take place during fertilisation ?
(A) Formation of secondary oocyte.
(B) Extrusion of IInd polar body from egg nucleus.
(C) Completion of Meiosis-II.
(D) Formation of haploid ovum.
Select the most appropriate answer from the options given below :

(1) (D) only

(2) (B) and (D) only

(3) (A), (C) and (D) only

(4) (B), (C) and (D) only

51) Given below are two statement :
Statement-I : After coitus, sperms are transported to the ampulla, where the sperm fertilises the ovum leading to formation of a diploid zygote.
Statement-II : The presence of X or Y chromosome in the sperm determines the sex of the embryo.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both statement-I and statement-II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect but statement-II is correct.

(4) Both statement-I and statement-II are correct.

52) Which of the following options in the given table correctly states about human egg ?

	Primary egg membrane	**Secondary egg membrane**	**Perivitelline space**
(1)	Zona pellucida	Corona radiata	Space between zona pellucida and corona radiata
(2)	Corona radiata	Zona pellucida	Space between plasma membrane of oocyte and zona pellucida
(3)	Zona pellucida	Corona radiata	Space between plasma membrane of oocyte and zona pellucida
(4)	Corona radiata	Zona pellucida	Space between corona radiata and zona pellucida

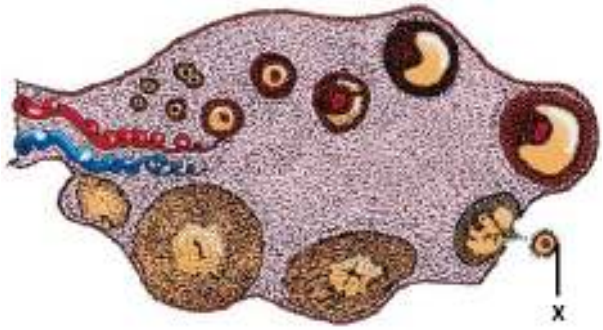
(1) 1

(2) 2

(3) 3

(4) 4

53) Structure marked as 'X' which is shown in the below diagram is –



(1) Haploid ovum (ootid) with 22+X chromosomes.

(2) Mature graffian follicle showing antrum.

(3) Secondary oocyte along with it's egg membranes.

(4) Corpus luteum.

54) Read the given statements :-
(a) Testes are maintained in environment with 2-2.5°C lower than normal body temp.
(b) Each testicular lobules contains 250 highly coiled tubules.
(c) Each seminiferous tubule is lined on outside by male germ cells.
Select correct option :-

(1) a and b are incorrect

(2) a and c are correct

(3) b and c are correct

(4) b and c are incorrect

55) Read the given statements and select option with incorrect statement :-

(1) Humans shows sexual dimorphism and reproductive events are similar in male and female.

(2) Some of reproductive events does not occur in male body while these occurs in female.

(3) Formation of ovum may continue in woman up to the age of fifty years.

(4) Events of gametogenesis are different in male and female.

56) Match the column - (I) with column (II) and select the correct option :-

	Column-I		**Column-II**
(a)	hCG	(i)	Placental hormone
(b)	Cleavage	(ii)	Birth hormone
(c)	Oxytocin	(iii)	Mitotic division

(d)	Amnion	(iv)	Extra embryonic membrane

(1) (a)-iv, (b)-iii, (c)-ii, (d)-i

(2) (a)-i, (b)-iii, (c)-iv, (d)-ii

(3) (a)-i, (b)-iii, (c)-ii, (d)-iv

(4) (a)-ii, (b)-iii, (c)-i, (d)-iv

57) Match the column (I) with column (II) and select the correct option :-

	Column-(I)		**Column - (II)**
(a)	Heart is formed	(i)	By end of 24 weeks
(b)	Eye lashes are formed	(ii)	During 5th month of foetal life
(c)	Hair on the head of foetus	(iii)	After one month of pregnancy
(d)	The foetus is fully developed	(iv)	By the end of 9 months

(1) (a) iv, (b) iii, (c) ii, (d) i

(2) (a) iii, (b) i, (c) iv, (d) ii

(3) (a) i, (b) iii, (c) iv, (d) ii

(4) (a) iii, (b) i, (c) ii, (d) iv

58) Which of the following statements are correct :
(A) Sertoli cells provide nutrition to the germ cells.
(B) The oviducts, uterus and vagina constitute the female accessory ducts.
(C) In early phase of pregnancy relaxin is secreted by the ovary.
(D) Signals for parturition originate from the fully developed foetus only
Options :

(1) A & C

(2) A & B

(3) B & C

(4) C & D

59)

Select the option that has the correct sequence of events of human female reproductive cycle.
(i) Secretion of FSH
(ii) Ovulation
(iii) Growth of corpus luteum
(iv) Growth of follicle and oogenesis
(v) Sudden increase in LH
Options :

(1) i-iii-v-iv-ii

(2) i-iv-v-ii-iii

(3) iv-i-v-iii-ii

(4) v-ii-iii-iv-i

60) Match the Column-I with II and choose correct option:

	Column-I		**Column-II**
(A)	Spermatogonia	(i)	n
(B)	Primary spermatocyte	(ii)	2n
(C)	Secondary spermatocyte		
(D)	Spermatid		
(E)	Sperm		

	(i)	**(ii)**
(1)	C, D	E, A, B
(2)	C, D, E	A, B
(3)	A, B	C, D, E
(4)	A, B, C	D, E

(1) 1

(2) 2

(3) 3

(4) 4

61) Choose that how many are correct among following statements-
(A) Seminal plasma with sperms is called semen.
(B) Male urethra is common passage for urine and semen.
(C) Presence or absence of hymen is reliable sign of virginity of female.
(D) For normal fertility of male at least 60% of ejaculated sperms must have normal size and shape.

(1) 4

(2) 1

(3) 2

(4) 3

62) How many of the following structures may not found in fallopian tube ?
Morula, secondary oocyte, primary oocyte, secondary follicle, ovum, zygote, blastocyst, gastrula

(1) 2

(2) 3

(3) 4

(4) 5

63) Which of the following included in the vulva?
(A) Mons Pubis
(B) Labia Majora
(C) Labia Minora
(D) Hymen
(E) Clitoris
(F) Prepuce
(G) Cervix
(H) Bartholin's gland
(I) Urethral Opening

(1) A, B, C, D, E
(2) A, B, C, D, E, F
(3) A, B, C, G, I
(4) A, B, C, E, G, H

64) Read the following statements (i to v) and answer the following question.
(i) Each testes has highly coiled 250 compartments called seminiferous tubules.
(ii) Erection of the penis due to presence of special tissues facilitates insemination
(iii) Immunologically competent cells are also present in the interstitial spaces out side the seminiferous tubules.
(iv) Testes lie outside the abdominal cavity in a pouch like skin called scrotum.
(v) Bulbourethral gland is a single accessory gland.
How many of the above statements are incorrect?

(1) (i), (ii) and (iii)
(2) (iii) and (v)
(3) (i) and (v)
(4) (ii), (iv) and (v)

65) Choose the incorrect statement from the followings :-

(1) High levels of estrogen triggers the ovulatory surge.
(2) Oogonial cells give rise to functional diploid ovum in regular cycle from puberty onwards.
(3) Sperms released from seminiferous tubules by the process called spermiation.
(4) Progesterone level is high during the post ovulatory phase of menstrual cycle.

66) Which of the following is incorrectly paired with its function ?

(1)	Ovary	Synthesis and secretion of steroid hormones.
(2)	Fimbriae	Collection of the ovum after ovulation
(3)	Seminal vesicle	Produces a sugar containing fluid to nourish sperm.
(4)	Bartholin glands	Secrete alkaline fluid to destroy the acidity of the urethra.

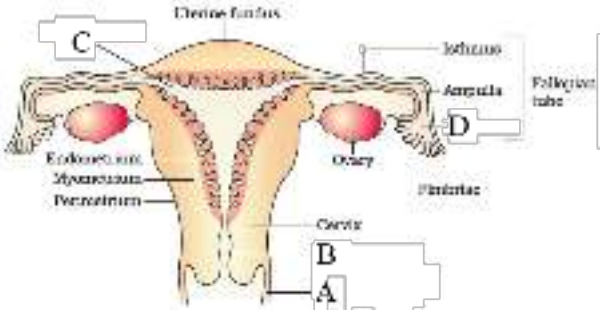
(1) 1

(2) 2

(3) 3

(4) 4

67) During copulation, semen is released by penis into which of the following labelled parts as A, B, C, D in the given diagram?



(1) A

(2) B

(3) C

(4) D

68) **Assertion :-** Cyclic menstruation is an indicator of normal reproductive phase.
Reason :- In human beings, menstrual cycles ceases around puberty age that is termed as menarche.

(1) Both assertion and reason are true and reason is the correct explanation of assertion

(2) Both assertion and reason are true but the reason is not the correct explanation of assertion

(3) Assertion is true but reason is false

(4) Both assertion and reason are false

69) **Assertion :** The male sex accessory ducts include rete testis, vasa efferentia, epididymis and vas deferens
Reason : The male accessory glands include paired seminal vesicles, a prostate and paired bulbourethral glands.

(1) Assertion and reason both are true and the reason is correct explanation of assertion

(2) Assertion and reason both are true but reason is not correct explanation of assertion

(3) Assertion is true but reason is wrong

(4) Assertion and reason both are wrong

70) **Statement-I :** The placenta is connected to the embryo through an umbilical cord which help in transport of substances.
Statement-II : Placenta secrets hCG, hPL only but not estrogen & progestogens.

(1) Statement-I & II both are correct.

(2) Statement-I & II both incorrect.

(3) Only statement-I correct.

(4) Only statement-II correct.

71) Match the following columns :-

Column-A		Column-B	
(a)	Cleavage	(i)	hPL
(b)	Fertilisation	(ii)	Fusion of ♂ & ♀ gamete
(c)	Placenta	(iii)	Blastomere
(d)	Morula	(iv)	Mulberry fruit

Select the correct option :-

(1) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

(2) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(4) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

72) **Assertion :-** All copulations do not lead to fertilization & pregnancy.
Reason :- Fertilization can only occur if ovum & sperm are transported simultaneously to ampulla.

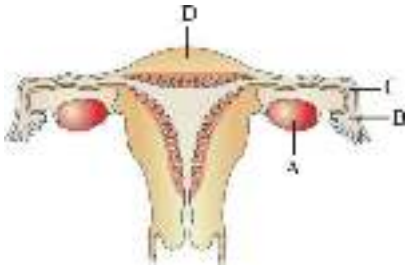
(1) Assertion is true, reason is false.

(2) Both Assertion and reason are true & reason is correct explanation of Assertion.

(3) Both Assertion and reason are true but reason is not correct explanation of Assertion.

(4) Both Assertion and reason are false.

73) In the given figure identify the parts labelled as A,B,C and D and select correct option :-



	A	B	C	D
(1)	Cervix	Infundibulum	Isthmus	Fundus
(2)	Ovary	Ampulla	Isthmus	Fimbriae
(3)	Ovary	Infundibulum	Ampulla	Fundus
(4)	Cervix	Isthmus	Ampulla	Fimbriae

(1) 1

(2) 2

(3) 3

(4) 4

74) A female who can produce ova and provide site for implantation and further development but

her life partner suffer from asthenozoospermia then best ART will be :

(1) ZIFT
(2) GIFT
(3) AI
(4) IUT

75) Which of the following is incorrectly matched pair ?

(1) Implant – Steroidal contraceptive which is inserted under the skin.
(2) Diaphragm– Make the uterus unsuitable for implantation and the cervix hostile to the sperm
(3) Multiload - 375 - Cu-releasing IUD
(4) Vaults – Reusable barrier device

76) Which contraceptive method is effective for preventing fertilization and protecting against transmission of sexually transmitted diseases ?

(1) Oral contraceptives
(2) Diaphragm
(3) Condom
(4) Intrauterine device (IUD)

77) How many statements are correct regarding Medical termination of Pregancy (MTP) ?
(a) Government of India legalised MTP in 1976 with some strict conditions to avoid its misuse.
(b) MTP are also essential in certain cases where continuation of the pregancy could be harmful or even fatal to mother or fetus or both.
(c) MTPs are considered relatively safe during the second trimester of pregancy.
(d) In first trimaster MTPs are much riskier.

(1) 4
(2) 3
(3) 2
(4) 1

78) Which of the following is a natural contraceptive method ?

(1) Vaults
(2) IUDs
(3) periodic abstinence
(4) Saheli

79) **Assertion (A) :** India initiated a nation wide family planning programme in 1951
Reason (R) : The aim was to reduce maternal and infant mortality rates only.

(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is not the correct explanation of A
(3) A is true but R is false

(4) A is false but R is true

80) Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion A : Amniocentesis for sex determination is one of the strategies of Reproductive and Child Health Care Programme.
Reason R : Ban on amniocentesis checks increasing menace of female foeticide.
In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **A** and **R** are true and **R** is NOT the correct explanation of **A**

(2) **A** is true but **R** is false

(3) **A** is false but **R** is true

(4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

81) Given below are two statements :-
Statement-I : Early symptoms of most of the sexually transmitted diseases are minor and include itching, fluid discharge, slight pain and swelling etc. in the genital region.
Statement-II : Infections or diseases which are transmitted through sexual intercourse are collectively called sexually transmitted infections or venereal diseases.
In the light of the above statements select the correct answer from the options given below-

(1) Statement-I and II are correct.

(2) Statement-I is incorrect but II is correct.

(3) Statement-I is correct but II is incorrect.

(4) Statement-I and II both are incorrect.

82) Read the following statements and choose the correct option :-
Statement A: MTP is performed to get rid of unwanted pregnancy.
Statement B: Hepatitis–B and HIV can be transmitted by using sterile needles and surgical instruments.

(1) Both the statements are incorrect

(2) Both the statements are correct

(3) Only statement A is correct

(4) Only statement B is correct

83) Match list-I with list-II

	List-I		**List-II**
A.	Genital herpes	I.	Treponema pallidum
B.	Gonorrhoea	II	Chlamydia sp.
C.	Chylamydiasis	III.	Neisseria sp.
D.	Syphilis	IV.	Herpes simplex virus

Choose correct option :-
(1) A-IV, B-III, C-I, D-II

(2) A-III, B-IV, C-I, D-II

(3) A-III, B-IV, C-II, D-I

(4) A-IV, B-III, C-II, D-I

84) **Assertion (A):** Copper-T acts as a contraceptive by increasing phagocytosis sperms.
Reason (R): Copper ions released by the device make the cervix hostile to the sperms.

(1) Both A and R true, R is correct explanation of A

(2) Both A and R true, R is not correct explanation of A.

(3) A true, R false

(4) A false, R true

85) In the given list how many could be the reasons for infertility ? Physical, congenital, diseases, drugs, immunological, Psychlogical.

(1) 4

(2) 6

(3) 5

(4) 3

86) How many of the following are characters of ideal contraceptives?
user-friendly, surgical, easily available, effective, breakthrough bleeding, reversible, least side-effects, interfere with the sexual desire.

(1) Seven

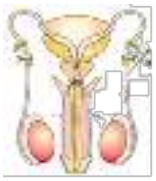
(2) Six

(3) Five

(4) Four

87) Identify the following structures and choose the correct option

(A)  (B) 

(C)  (D) 

1.	(A) Male condom	Cover the penis in the male just before coitus
	(B) Cu T	Make the uterus unsuitable for implantation
2.	(A) Male condom	Cover the vagina in the male just before coitus
	(D) Tubectomy	Alter the quality of cervical mucus to prevent/retard entry of sperms

3.	(B) Cu releasing IUD	Decreases sperm motility and fertilizing capacity
	(C) Implant	Inhibits ovulation
4.	(C) Implant	Inhibit ovulation and implantation
	(D) Vasectomy	small part of the vas deferens is removed or tied up through a small incision on the abdomen

(1) 1

(2) 2

(3) 3

(4) 4

88) Given below are two statements :-
Statement-I : STD's are a major threat to a healthy society.
Statement-II : Though all persons are vulnerable to these infections, their incidences are reported to be very high among persons in the age group of 45 to 60 years.
Choose the correct answer from the options given below :-

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

89) Which assisted reproductive technique is used for female infertility due to blocked oviducts?

(1) ICSI

(2) ZIFT

(3) IVF-IUT

(4) Both 2 and 3

90) In IVF and ET technique A or early embryo B could be transferred into fallopian tube by the method called C.
In above statements, identify the A, B and C :

	A	**B**	**C**
(1)	Sperm	Ovum	GIFT
(2)	Ovum	Zygote	ZIFT
(3)	Gamete	Upto 16 blastomeres	IUT
(4)	Zygote	upto 8 blastomeres	ZIFT

(1) 1

(2) 2

(3) 3

(4) 4

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	2	2	1	1	1	3	4	3	4	2	2	4	2	2	4	1	3	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	1	2	2	3	3	2	1	4	2	4	2	2	1	2	4	2	2	2	1
Q.	41	42	43	44	45															
A.	1	2	1	4	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	3	3	4	4	2	3	2	4	2	2	2	3	3	4	4	4	3	3	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	3	3	4	1	3	2	2	3	3	4	4	2	4	2	4	2	1	3	2
Q.	86	87	88	89	90															
A.	3	4	2	2	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	3	2	2	1	1	4	1	2	3	4	4	4	2	2	2	3	2	2	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	3	3	3	2	3	1	4	1	2	3	2	3	3	4	4	2	2	2	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	1	3	2	4	1	3	3	3	4	4	3	3	4	1	3	4	2	2	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	3	1	3	2	4	1	3	2	3	1	2	3	3	2	3	4	3	3	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	3	4	3	2	3	3	3	3	4										

SOLUTIONS

PHYSICS

1) $V_0 = \frac{kQ}{R}\ V_A = \frac{kQ}{6R}$

$w_{ext} = -Q[V_0 - V_A]$

$= -Q\left[\frac{kQ}{R} - \frac{kQ}{6R}\right] = -\frac{5kQ^2}{6R}$

2) Electric field at surface $E = \frac{kQ}{R^2}$

$\Rightarrow Q = \frac{ER^2}{k} \Rightarrow ne = \frac{ER^2}{k}$

$\Rightarrow n = \frac{ER^2}{ke}$

$= \frac{0.36 \times (0.1)^2}{9 \times 10^9 \times 1.6 \times 10^{-19}}$

$= 2.5 \times 10^6$

3)

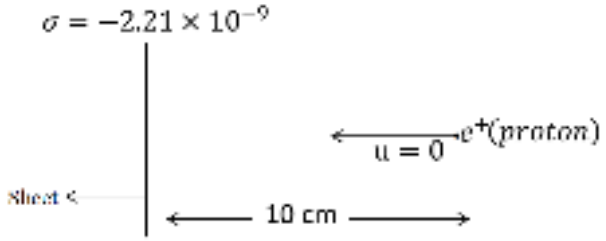
in (x-y) plane distance r from origin is given by

$r = \sqrt{x^2 + y^2}$

because, $V = -Kxy$

$\Rightarrow -\frac{\partial V}{\partial x} = E_x = Ky$ and $-\frac{\partial V}{\partial y} = E_y = Kx$

$E = \sqrt{E_x^2 + E_y^2} = K\sqrt{x^2 + y^2} = Kr \Rightarrow \boxed{E \propto r}$



4)

$u = 0,\ F = qE$ [F = force on proton]

$F = e^+E$ [$q = e^+$ = charge of proton $= 1.6 \times 10^{-19}$ C]

$a = \frac{e^+E}{m_p}$ [$\because F = m_pa$, m_p = mass of proton $= 1.67 \times 10^{-27}$kg, a = acceleration of proton]

By the equation, $s = ut + \frac{1}{2}at^2$

$t^2 = \frac{2s}{a}$ [∵ u = 0 , s = 10×10^{-2}m]

$\Rightarrow t^2 = \frac{2s}{\frac{e^+E}{m_p}}$

Electric field due to infinite charged sheet is given by $E = \frac{\sigma}{2\varepsilon_0}$

Putting the value of 'E'

$\Rightarrow t^2 = \frac{2s}{e^+ \times \sigma} \times m_p \times 2\varepsilon_0$

Putting the values, we get

$t^2 = \frac{2 \times 10 \times 10^{-2} \times 1.67 \times 10^{-27} \times 2 \times 8.85 \times 10^{-12}}{1.6 \times 10^{-19} \times 2.21 \times 10^{-9}}$

$\Rightarrow t^2 = 16.71 \times 10^{-12}$

$t \approx 4 \times 10^{-6} \approx 4\mu s$

5) $\frac{\Delta V}{\Delta r} = E \Rightarrow \Delta V = \frac{\sigma}{2\varepsilon_0}.\Delta r$

$50 = \frac{0.1 \times 10^{-6}}{2 \times 8.85 \times 10^{-12}} \Delta r$

$\Rightarrow \Delta r = 8.85 \times 10^{-3}$ m

$\Delta r = 8.8$ mm

6) • **Case P (All positive):** All forces push charge q away. Net force is upward (+Y) Match with c.

• **Case Q (Q_1, Q_2 positive, Q_3, Q_4 negative):** Forces from Q_1, Q_2 are repulsive, and those from Q_3, Q_4 are attractive. Net force is toward + x Match with a.

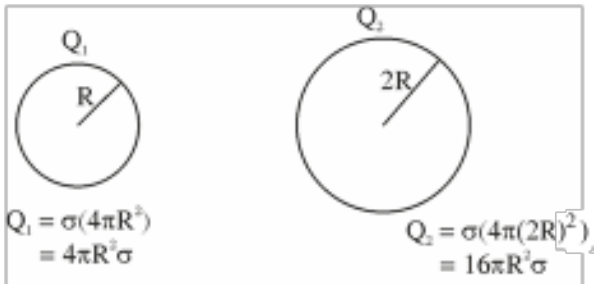
• **Case R (Q_1 , Q_4 positive, Q_2, Q_3 negative):** The forces cancel in x-direction, leaving net force in – y Match with d.

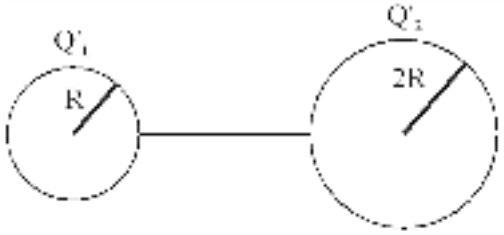
• **Case S (Q_1, Q_3 positive, Q_2, Q_4 negative):** Net force is directed – x Match → with b.

7) $q = \int_0^R \rho dV = \int_0^R \rho_0 \left(\frac{5}{4} - \frac{r}{R}\right) 4\pi r^2 dr = \frac{2\pi\rho_0 R^3}{3}$

At surface

$V = \frac{1}{4\pi\epsilon_0}\frac{q}{R}, = \frac{1}{4\pi\epsilon_0 R} \times \frac{2\pi\rho_0 R^3}{3} = \frac{\rho_0 R^2}{6\epsilon_0}$

8) 



as $V_1' = V_2'$

$$\frac{Q'_1}{4\pi\varepsilon_0 R} = \frac{Q'_2}{4\pi\varepsilon_0(2R)}$$

$$Q'_2 = 2Q'_1$$

$$Q'_1 + Q'_2 = Q_1 + Q_2$$

$$\frac{Q'_2}{2} + Q'_2 = 20\pi R^2\sigma$$

$$\frac{3}{2}Q'_2 = 20\pi R^2\sigma$$

now σ' = $$\frac{Q'_2}{4\pi(2R)^2} = \frac{2}{3}\cdot\frac{20\pi R^2\sigma}{16\pi R^2} \Rightarrow \frac{\sigma'}{\sigma} = \frac{5}{6}$$

9) $$E = \frac{k\lambda}{a}(\sin\theta_1 + \sin\theta_2)$$

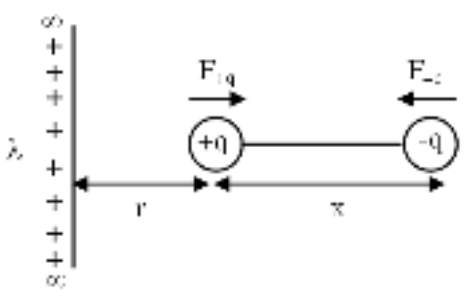
$$E = \frac{1}{4\pi\varepsilon_0}\times\frac{Q}{L}\times\frac{1}{\left(\frac{\sqrt{3}L}{2}\right)}\times(2\sin\theta)$$

$$\tan\theta = \frac{L/2}{\frac{\sqrt{3}L}{2}} = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30°$$

$$\sin\theta = \frac{1}{2}$$

$$E = \frac{1}{4\pi\varepsilon_0}\times\frac{2Q}{\sqrt{3}L^2}\times\left(2\times\frac{1}{2}\right)$$

$$E = \frac{Q}{2\sqrt{3}\pi\varepsilon_0 L^2}$$

10) 

r = 10 mm, x = 2mm

$$\left|\vec{F}_q\right| = \frac{2k\lambda}{r}.q$$

$$\left|\vec{F}_{-q}\right| = \frac{2k\lambda}{r+x}.q$$

$$\Rightarrow \left|\vec{F}_{net}\right| = \frac{2k\lambda q}{r} - \frac{2k\lambda q}{r+x}$$

$$\left|\vec{F}_{net}\right| = \frac{2k\lambda q.x}{r(r+x)}$$

$$4 = \frac{2 \times 9 \times 10^9 \times 3 \times 10^{-6} \times q \times 2mm}{10mm.12mm}$$

$\Rightarrow q = 4.44\ \mu C$

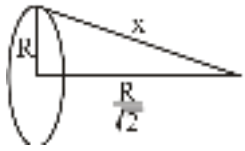
11)

$$E_{rem} = E_{total} - E_{use}$$

$$= \frac{k\,q}{(2R)^2} - E$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{\rho \times \frac{4}{3}\pi R^3}{4R^2} - E$$

$$= \frac{\rho R}{12\epsilon_0} - E$$

12) 

$$x = \sqrt{R^2 + \frac{R^2}{2}} = \sqrt{\frac{3}{2}}R$$

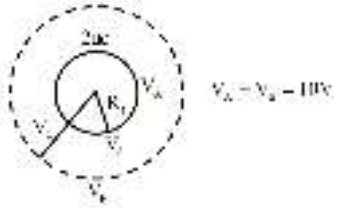
13)

$$v = \sqrt{\frac{2qV}{m}} = \sqrt{\frac{2qE.d}{m}}$$

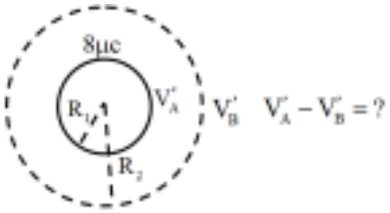
$$v = 4\sqrt{2} \times 10^5$$

14)

Case (1)



Case (2)



For case (1)

$$V_A - V_B = \frac{k(2\mu C)}{R_1} - \frac{k(2\mu C)}{R_2} \quad \text{----- (1)}$$

For case (2)

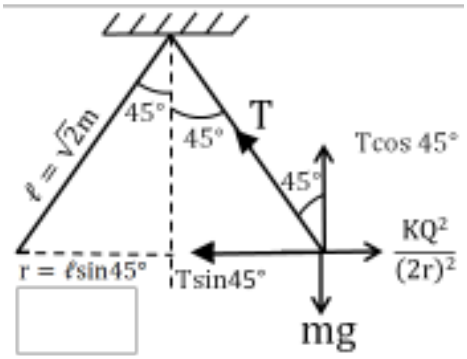
$$V_A' - V_B' = \frac{k(8\mu C)}{R_1} - \frac{k(8\mu C)}{R_2}$$

$$V_A' - V_B' = 4\left[\frac{k(2\mu C)}{R_1} - \frac{k(2\mu C)}{R_1}\right]$$

$$V_A' - V_B' = 4\,[V_A - V_B]$$

$$V_A' - V_B' = 4 \times 10 = 40V$$

15)



from the diagram.

$$T\sin 45^o = \frac{KQ^2}{4r^2} \quad ...(1)$$

$$T\cos 45^\circ = mg \quad ...(2)$$

eq (1) /eq (2)

$$\tan 45^\circ = \frac{\frac{KQ^2}{4r^2}}{mg}$$

$$mg = \frac{KQ^2}{4r^2}$$

$$m = \frac{9 \times 10^9 \times 10 \times 10^{-6} \times 10 \times 10^{-6}}{4 \times (1)^2 \times 10}$$

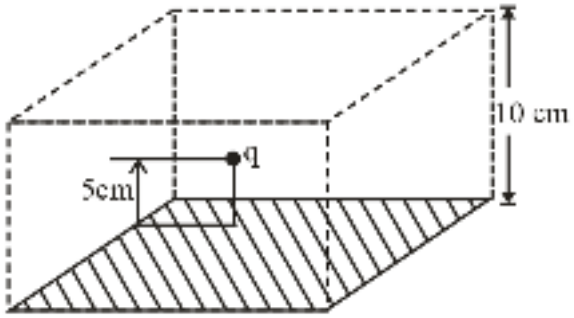
$$m = \frac{9}{4} \times 10^{-2} kg$$

$$m = 22.5\ g$$

16)

$$\phi \text{ shaded surface} = \frac{q}{6\varepsilon_0}$$

$$\phi = \frac{10 \times 10^{-6}}{6 \times 8.85 \times 10^{-12}} = 1.88 \times 10^5\ Nm^2/c$$



17)

No electric field line exist in conductor and electric field lines originate or terminate perpendicularly on surface of conductor.

18)

Potential difference = $\frac{KQ}{r_1} - \frac{KQ}{r_2}$

$r_1 = \sqrt{\left(\sqrt{3}\right)^2 + \left(\sqrt{3}\right)^2}$

$r_2 = \sqrt{\left(\sqrt{6}\right)^2 + 0}$

As $r_1 = r_2 = \sqrt{6}m$

So potential difference = 0

19)

Part-1

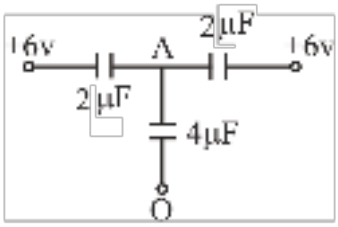
$$U_i = U_1 + U_2 = \frac{1}{2}CV^2 + \frac{1}{2}CV^2 = CV^2$$

$$U_f = KU_1 + \frac{U_1}{K} = \frac{3CV^2}{2} + \frac{CV^2}{2 \times 3}$$

$$= \frac{10CV^2}{6}$$

$$= \frac{5}{3}CV^2$$

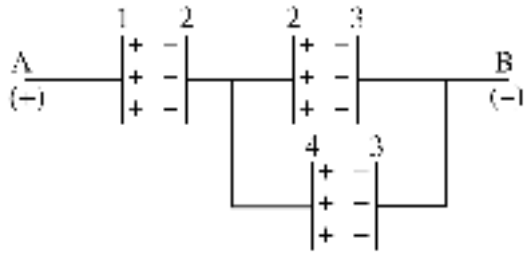
$$\frac{U_i}{U_f} = \frac{CV^2}{\frac{5}{3}CV^2} = \frac{3}{5}$$

20) 

Potential at A $= \frac{C_1V_1 + C_2V_2 + C_3V_3}{C_1 + C_2 + C_3}$

= 3 volt

 Charge of $C_1 = Q_1 = C_1 (6 - 3)$ $= 2\mu F \times 3 = 6\ \mu C$



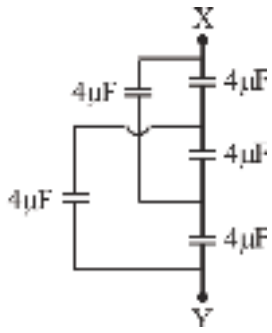
21)

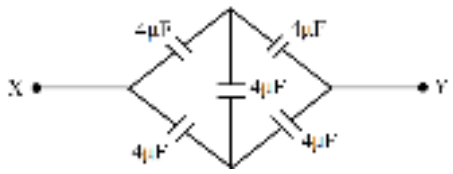
$C_{eq} = \frac{2C}{3} = \frac{2}{3}\frac{\epsilon_0 A}{d}$

$Q = C_{eq}E = \frac{2}{3}\frac{\epsilon_0 AE}{d}$

22) Q = CV
At steady state
V = 2V
Q = (10 μF) (2)
Q = 20 μC

23)





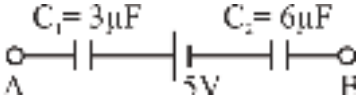
⇒
$C_T = 4\mu F$

24)

Initially, capacitance is $C = \frac{K\epsilon_0 A}{d}$. If the distance is halved, $d' = \frac{d}{2}$, and the dielectric constant is tripled, $K' = 3K$, then the new capacitance is : C' = 6C
This shows that the capacitance becomes six times the original value.

25)

$V_A - V_B = 7V$



By KVL

$$V_A - \frac{q}{3} - 5 - \frac{q}{6} - V_B = 0$$

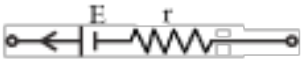
$$q\left[\frac{3}{6}\right] = 2$$

q = 4μC

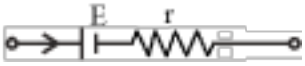
$$U_{C_1} = \frac{q^2}{2C} = \frac{16}{2 \times 3} = \frac{8}{3}\mu J$$

26)

6 parallel branches having 3 capacitors of 3 μF, 400 V in series. Total no. of capacitors
N = 6 × 3 = 18

27) 

T.P.D = E – I r ⇒ 12 = E – 2r ...(1)



T.P.D = E + I r ⇒ 15 = E + 3r ...(2)
Solve eqn (1) & (2)

28)

In both loops of circuit current I_1 and I_2 are

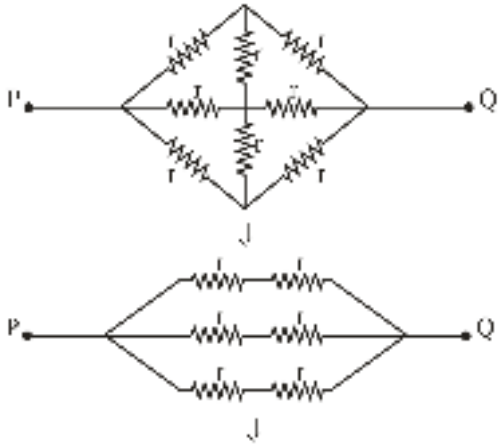
$I_1 = \frac{10}{5} = 2A$

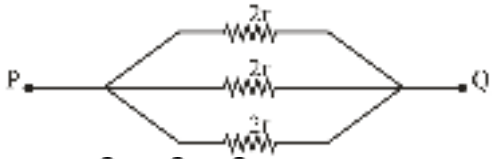
$I_2 = \frac{20}{10} = 2A$

Hence no current will flow from 2Ω resistor

29) In series $20 = \frac{P}{n}$; In parallel Pn = 2000
⇒ P = 200 & n = 10

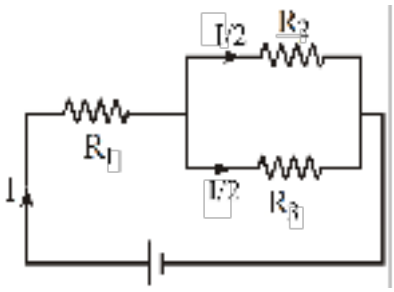
30) Using symetric line method
The given circuit can be simplified as follows





$$R_{PQ} = \frac{2r}{3} = \frac{2}{3} \times \frac{3}{2} = 1\Omega$$

31) $i = \vec{J} \cdot \vec{A}$
$= (2 + 12) \times 10^{-6}A$
$= 14\ \mu A$



32)
Since All resistance are equal
Power in R_1 will be I^2R_1 which will be greatest.
$P_1 = I^2R$

$$P_2 = P_3 = \left(\frac{I}{2}\right)^2 R = \frac{I^2R}{4}$$

33) Under balance bridge condition, then on interchanging the position of galvanometer and cell, condition of balance does not change.

34)

Slope of VI curve = R = tanθ
$(Slope)_1 > (Slope)_2 \Rightarrow R_1 > R_2$
for conductor if T↑ Then R↑
so $T_1 > T_2$

35) The formula of drift velocity is $v_d = \frac{eE}{m}\tau$
Current density $J = \frac{I}{A} = \frac{neAv_d}{A} = nev_d$
Relaxation period $\tau = \frac{m}{ne^2\rho}$
Resistance is $R = \frac{V}{I}$

$$\rho\frac{\ell}{A} = \frac{E\ell}{I} \Rightarrow \rho = \frac{EA}{I} = \frac{E}{J}$$

36) $I = \frac{30}{\frac{12\times6}{18} + R} = \frac{30}{4+R} = 5$

⇒ R = 2Ω

37)

Inverse of slope of i–V curve is resistance.

38) $\frac{X}{80} = \frac{Y}{20} \Rightarrow X = 4Y$ and $\frac{X'}{50} = \frac{Y}{50} \Rightarrow X' = Y$

$X' = \frac{10X}{10+X} = Y = \frac{X}{4} \Rightarrow 40 = 10 + X$

X = 30 Ω & Y = 7.5 Ω

39) For parallel coils

$\frac{1}{t_{eff.}} = \frac{1}{t_1} + \frac{1}{t_2} \Rightarrow t_{eff} = \frac{10 \times 15}{10+15} = 6$ min.

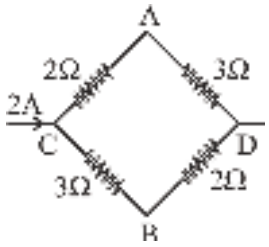
40) प्रत्येक भाग का प्रतिरोध r = R/10

माना तुल्य प्रतिरोध r_R है तब

$\frac{1}{r_R} = \frac{1}{r} + \frac{1}{r} +$ .....

$r_R = \frac{R}{100} = 0.01\ R$

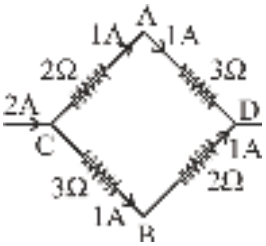
41)



Resistance of the upper arm CAD = 2Ω + 3Ω = 5 Ω

Resistance of the lower arm CBD = 3Ω + 2Ω = 5Ω

As the resistance of both arms are equal, therefore same amount of current flows in both the arms.



Current through each arm CAD or CBD = 1A

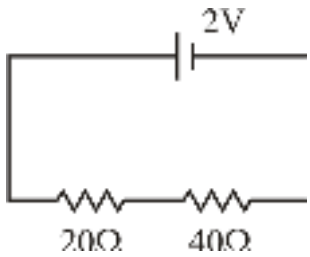
Potential differences across C and A is

$V_C - V_A = (2\ \Omega)\ (1\ A) = 2\ V$ ...(i)

Potential differences across C and B is

$V_C - V_B = (3\ \Omega)\ (1\ A) = 3\ V$ ...(ii) Subtracting (i) from (ii), we get $V_A - V_B = 3\ V - 2\ V$

= 1 V



42)

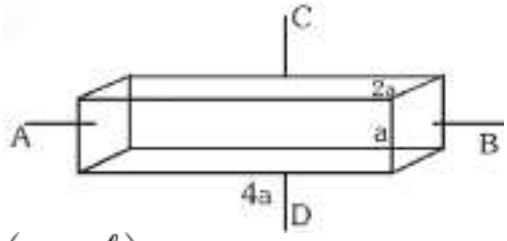
$i = \frac{2}{60}$

$\Delta V = i \times 40 = \frac{2}{60} \times 40 = \frac{4}{3} = 1.33$ V

43)

Maximum power delivered when internal resistance equals external resistance.
$\therefore R = 0.5\,\Omega$

44) x > z > y



$\left(R = \rho\frac{\ell}{A}\right)$, A = Area of cross-section.

45)

Apply KVL from A to B
V_A – (3) (2) – 3 – (4) (1) + 2 – 6 (1) = V_B
$V_A - V_B$ = 17 Volt

CHEMISTRY

46)

C.P. ∝ i × m
0.2 m Glucose (i = 1) = 0.2 Lowest C.P.
0.1 m Na_3PO_4 (i = 4) = 0.4 Highest C.P.
0.05 m $Al_2(SO_4)_3$ (i = 5) = 0.25
O.P., B.P. ∝ C.P.
V.P., $F.P \propto \frac{1}{C.P.}$

47) $\lambda_N = \lambda_{M/v.F.} = \lambda_{M/2} = \lambda_M = 200$

$$\lambda_M = \frac{K \times 1000}{S}$$

$$200 = \frac{10^{-6} \times 1000}{S}$$

$\Rightarrow S = 5 \times 10^{-6}$

$Ag_2CrO_4 \rightleftharpoons 2Ag^+ + CrO_4^{2-}$

S 2S S

$K_{sp} = 4S^3 = 5 \times 10^{-16}$

48)

Ca ⇒ Cannot be reduced only oxidised.

49) $\Delta T_b = iK_bm$

$$i = \frac{\Delta T_b}{K_b \times m} = \frac{1.5}{2.5 \times 1} = 0.6$$

$$i = 1 - \beta + \frac{\beta}{n}$$

$$0.6 = 1 - \beta + \frac{\beta}{2} = 1 - \frac{\beta}{2}$$

$$\frac{\beta}{2} = 0.4$$

β = 0.8 so 80%

50) NCERT

51) $\frac{k_1}{k_2} = \frac{R_2}{R_1}$

$k_2 = 0.4$

$$\lambda_m = \frac{k}{1000M} = \frac{0.4}{1000 \times 0.4} = 10^{-3}$$

52) Elementary A → 2B first order

t = 0 2.5 0

t = 35 2.5 – 1 2 [40% reaction]

= 1.5

$$= \frac{1}{2.5} \times 100 = 40\%$$

$$\frac{2.3}{35} \log \frac{2.5}{1.5} = \frac{2.3}{t} \log \frac{8}{4.8}$$

A → 2B

8 – x 2x = 6.4

8 – 3.2 x = 3.2

= 40% reaction

For 1st order reaction same % of reaction same time

53) $P_{CO_2} = K_H \times \frac{n}{N}$

$n = P_{CO_2} \times \frac{N}{K_H}$

$= 2.5 \times \frac{1500}{18} \times \frac{1.01 \times 10^5}{1.67 \times 10^8} = 0.126$ mol

Mass = 5.56 g

54) $\Delta T_f = i\, k_f m$

$= 2 \times 1.86 \times \frac{24}{40} \times \frac{1000}{372} = 6$

F.P. = –6°C

So on cooling to –4°C no freezing

55) On dilution number of ions remain same but ionic mobility increases.

56)

	2A	→	3B
	a		0
	a – 2x		3x

at intersection a – 2x = 3x

a = 5x

$x = \frac{a}{5}$

$2x = \frac{2a}{5}$ or 40%

57) $\text{R.L.V.P.} = \frac{P^o - P}{P^o} = \frac{n}{N}$ (n → same)

$\text{R.L.V.P.} \propto n = \frac{W_B}{M_B}$

$\frac{(\text{R.L.V.P.})_A}{(\text{R.L.V.P.})_B} = \frac{M_A}{M_B}$

$\frac{1}{3} = \frac{M_A}{M_B}$

$M_B = 3M_A$

58) $\frac{\left(\log \frac{k_2}{k_1}\right)_A}{\left(\log \frac{k_2}{k_1}\right)_B} = \frac{\frac{E_A}{2.303R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)}{\frac{E_B}{2.303R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)}$

$\frac{\log 6}{\log 6^3} = \frac{E_A}{E_B}$

$\frac{\log 6}{3 \log 6} = \frac{E_A}{E_B}$

$\Rightarrow E_B = 3E_A$

59)

In dilute NaCl electrolysis of H_2O occurs.

60)

	A	→	3B
t = 0	P_0		
After same time	$P_0 - x$		3x
Very long time	0		$3P_0$

$P_t = P^0 + 2x \Rightarrow x = \frac{P_t - P_0}{2}$

$$K = \frac{1}{t}\ell n\frac{P_0}{P_0 - x} = \frac{1}{t}\ell n\frac{\frac{P_\infty}{3}}{\frac{P_\infty}{3} - \left(\frac{P_t - P_0}{2}\right)}$$

$$= \frac{1}{t}\ell n\left[\frac{\frac{P_\infty}{3}}{\frac{P_\infty}{3} - \frac{(3P_t - P_\infty)}{6}}\right]$$

$$K = \frac{1}{t}\ell n\left(\frac{P_\infty/3}{\frac{P_\infty}{2} - \frac{P_t}{2}}\right)$$

61) $n_A = \frac{1}{2}, \quad n_B = \frac{1}{4}$

$x_A = \frac{2}{3}, \; x_B = \frac{1}{3}$

$P_T = P_A^o x_A + P_B^o x_B = 80$ torr

$y_B = \frac{P_B^o x_B}{P_T} = \frac{40}{80} \times \frac{1}{3} = \frac{1}{6}$

62) $P_{ext} < P_{th}$

 –ve deviation

63)

Order of a reaction can have fractional value

64) pH = 12 $[H^+] = 10^{-12}$ $[OH^-] = 10^{-2}$

$[Cu^{2+}][OH^-]^2 = K_{sp}$

$Cu^{2+} = \frac{K_{sp}}{[OH^-]^2} = \frac{10^{-19}}{10^{-4}} = 10^{-15}$

$Cu^{2+} + 2e^- \rightarrow Cu$

$$E = E^o_{cell} + \frac{0.0591}{2}\log[Cu^{2+}]$$

$E_{cell} = +0.34 + \frac{0.0591}{2}\log 10^{-15}$

$E_{cell} = -0.11$ V

65) Eq. of H_2O_2 = Eq. of $KMnO_4$

$2 \times 40 \times M_{H_2O_2} = 5 \times 0.2 \times 20$

$$M_{H_2O_2} = \frac{5 \times 0.2 \times 20}{2 \times 40}$$
$= 0.25$ M
Volume strength = M × 11.2
= 0.25 × 11.2
= 2.8

66) For zero order, $k = \frac{a}{2(t_{1/2})_A}$
For first order, $k = \frac{0.693}{(t_{1/2})_B}$
$$\frac{a}{2(t_{1/2})_A} = \frac{0.693}{(t_{1/2})_B}$$
$$\frac{0.693}{2(t_{1/2})_A} = \frac{0.693}{(t_{1/2})_B} \Rightarrow 2(t_{1/2})_A = (t_{1/2})_B$$

67) More active metal cannot be used as vessel for storage of solution of less metal salt.
⇒ Lower SRP means more active metal.

68) It is zero order reaction

69) $\pi \propto i \times M$
$$M = \frac{N}{V.F}$$
$i \times C$
NaCl = 2 × 1
$K_2SO_4 = 3 \times \frac{1}{2}$
$Na_3PO_4 = 4 \times \frac{1}{3}$
$K_2SO_4 = 3 \times 1$

70) $W_{solvent} = W_{solution} - W_{solute}$
= 1836 – 36 = 1800 g
$$m = \frac{n}{W} = \frac{36}{180} \times \frac{1000}{1800} = 0.111$$

71) $K = Ae^{-E_a/RT}$
$\ln K = \ln A - \frac{E_a}{RT}$
$2.303 \log K = 2.303 \log A - \frac{E_a}{RT}$
$\log K = \log A - \frac{E_a}{2.303R}\frac{1}{T}$
$\log K = -(1500)\frac{1}{T} + 7$
On comparing

$$\frac{-E_a}{2.303 \times R} \times \frac{1}{T} = \frac{-1500}{T}; \qquad \log A = 7$$

$E_a = 1500 \times 8.314 \times 2.303$ $\quad A = 10^7$

$= 28.7$ kJ/mol

72) $E_{cell} = E_{cathode} - E_{anode} = +ve$

(2nd eq.) (1st eq)

Mn^{2+} & MnO_4^- react to form MnO_2

73)

2nd order

74)

Iodine-starch (blue) when react with hypoblie colour disappear.

75)

NaOH + Phenolphthalein

↓ Pink Colour

↓ HCl

Colourless

76)

Lyophillic (Reversible) Biomolecule

Lyophobic irreversible inorganic salt

77) $$M = \frac{x(\%w/w) \times d \times 10}{mol.\ wt}$$

$$x = \frac{M \times mol.\ wt}{d \times 10} = \frac{3 \times 158}{1.58 \times 10} = 30$$

78) 0.8 to 0.1

$x \xrightarrow[21\ min]{3t_{1/2}} x/8$ means $t_{1/2} = 7$ min

$$K = \frac{0.693}{7} = 0.099$$

Rate = K(A) = $0.099 \times 0.05 = 4.95 \times 10^{-3}$

79) $Cl^{\ominus}$ can not act as O.A.

80) Eq. of $K_2Cr_2O_7$ = Eq. of FAS

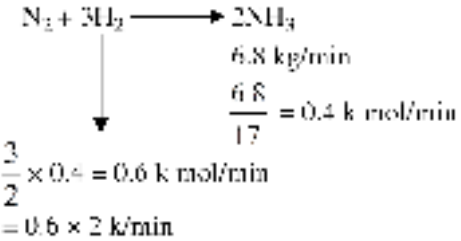
$6 \times 0.5 \times V_{consumed} = 1 \times 0.3 \times 25$

$V_{consumed} = 2.5$
$V_{remain} = 47.5$

81) $\pi = \frac{w}{v}\frac{RT}{M}$

$\log \pi = \log\left(\frac{w}{v}\right) + \log\left(\frac{RT}{M}\right)$

$\frac{RT}{M} = x \times 10^{-2}$



82) = 1.2 k/min

83) $k_{eq} = 10^{\frac{nE^o_{cell}}{0.0591}}$

$10^{12.7} = 10^{\frac{nE^o_{cell}}{0.0591}}$

$12.7 = \frac{nE^o_{cell}}{0.0591} = \frac{2 \times E^o_{cell}}{0.0591}$

$E^o_{cell} = 0.375 = E^o_{cathode} - E^o_{anode}$

$0.375 = E^o_{c^+/c} - E^o_{A^{2+}/A}$

$E^o_{c^+/c} = 0.375 + 0.75$

$= 1.125$

84)

Names of instruments

85) $N_1V_1 = N_2V_2$

$2 \times M \times 20 = 1 \times 0.5 \times 40$

$M = 5$

86) $n_{CCl_4} = \frac{77}{154} = 0.5$

$n_{CH_2Cl_2} = \frac{85}{85} = 1$

$P_T = P^o_A X_A + P^o_B X_B$

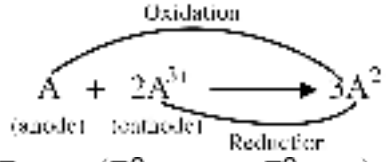
$= 150 \times \frac{1}{3} + 420 \times \frac{2}{3}$

$= 50 + 280 = 330$

87) $$ROR = \left[-\frac{dA}{dt}\right] = \frac{1}{3}\left[-\frac{dB}{dt}\right] = +\frac{1}{2}\left[\frac{dC}{dt}\right]$$

↓ ROD of A ↓ ROD of B ↓ ROD of C

88) For complete cell reaction



$$E_{cell} = (E^{o}_{Cathode} - E^{o}_{Anode})_{SRP}$$

E_{cell} = 0.84 – (–0.54)

= 1.38 V

89) $(\Delta H_{sol^n})_{Anhydrous\ Salt} = (\Delta H_{sol^n})_{Hydrated\ salt} + \Delta H_{Hydration}$

90) excess $AgNO_3$ so +ve charge on colloid

BIOLOGY

91) NCERT-XII, Pg. # 15

92) NCERT-XII, Pg. # 7, 10

93) NCERT-XII, Pg. # 18

94) NCERT XII Pg. # 13

95) NCERT-XII, Pg. # 13

96) NCERT-XII, Pg. # 10

97) NCERT-XII, Pg. # 18

98) NCERT-XII, Pg. # 6, 10, 18

99) NCERT-XII, Pg. # 7,8, 9

100) NCERT, Pg # 8,15,18, 19

101) NCERT, Pg. # 15

102) NCERT, Pg. # 10

103) NCERT XII Pg. # 28

104) NCERT-XII Pg. # 23,25

105)

NCERT-XII, Pg. # 5,6,9,10,11

106) NCERT Pg. # 19,20

107) NCERT Pg.#7,9.

108) NCERT-XII, Pg # 20,21,22

109) NCERT XIIth Pg.# 16,18,19,21.

110) NCERT-XII, Pg # 6, 7

111) NCERT Pg.# XIIth -20.

112) NCERT-XI, Pg. # 76

113)

NCERT-XI, Pg. # 76

114)

NCERT-XI, Pg. # 91

115) NCERT Pg. # 72

116) NCERT-XI, Pg. # 91

117) NCERT-XI, Pg. # 75

118)

NCERT, Pg. # 72

119) NCERT Pg. # 76,77

120) NCERT-XI, Pg. # 71, 72

121) NCERT, Pg. # 76

122)

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123) NCERT-XI Pg. No. # 69

124) NCERT-XII, Pg. # 63

125) NCERT Pg. 64

126) NCERT, Pg. # 64

127) NCERT Pg. # 63

128) NCERT Pg. No. # 65

129)

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130)

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131) NCERT Pg. # 62

132) XI-NCERT Page No. # 66

133) NCERT-XI, Pg. # 65

134) NCERT Pg # 58, 61, 62

135) NCERT Pg. # 75

136) NCERT Pg. # 34

137) NCERT Pg. # 38

138) NCERT Pg. # 38

139) NCERT Pg.# Module.

140) NCERT Pg.# English- 36.
Hindi- 39.

141) NCERT Pg.# English - 36.
Hindi - 39.

142) NCERT Pg.# Module.

143) NCERT Pg.#33-(English), (Fig. 2.7).

144) NCERT Pg. # 27

145) NCERT Pg. 26

146) NCERT Pg. 35-39

147) NCERT Pg. 38

148) NCERT, Pg # 27,29,37,38

149)

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150) NCERT-XII, Page No. 33

151) NCERT-XI, 28,29,30

152)

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153) NCERT Pg. # 30

154)

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155)

NCERT-XII, Pg. # 50, 51

156)

NCERT Pg. # 29

157)

NCERT Pg. # 29

158) NCERT-XII, Pg # 51

159)

NCERT Pg. # 28

160) NCERT Pg.# Module.

161) NCERT Pg.#36.

162) NCERT Pg.#35.

163) NCERT-P.No. 45

164)

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165)

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166)

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167)

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168) NCERT Pg. # 44

169) NCERT (XII) ; Page No. # 41, 42

170)

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171) NCERT Pg. # 47

172) NCERT PAGE (ENGLISH): 46,47

173) NCERT Pg. # 47

174) NCERT Pg. # 45

175) NCERT Pg. No. # 48

176) NCERT XII Pg. No. # 43

177) NCERT PAGE (ENGLISH): 44,45

178) NCERT Pg. # 47

179)

NCERT Pg. # 48

180) NCERT P.g No. 48